

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414863
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Hayes; Stanley Kyle et al.

Expandable interbody implant and corresponding surgical tool

Abstract

An interbody system including an implant and a tool for inserting and expanding the medical implant and locking the implant in place is disclosed. The medical implant may include an expandable body defined by a superior endplate and an inferior endplate that are hingedly coupled and may be expanded and lordosed. The superior and inferior endplate may include radially disposed and opposed surfaces that mate and/or directly contact each other when a locking screw is threaded through a screw aperture. The implant may include a threaded breakoff screw disposed in the threaded screw aperture and movable between a locked position and an unlocked position, for example. In the locked position, the threaded locking screw may urge the distal engagement surface of the first core into direct contact with the proximal engagement surface of the second core. When broken, the breakoff screw may comprise a recessed fracture surface.

Inventors: Hayes; Stanley Kyle (Mission Viejo, CA), Loke; Robert M. (Memphis, TN), Barfield; Charlie J. (East Hernando, MS), Protopsaltis; Dimitri K. (Memphis, TN)

Applicant: Warsaw Orthopedic, Inc. (Warsaw, IN)

Family ID: 1000008819710

Assignee: WARSAW ORTHOPEDIC, INC. (Warsaw, IN)

Appl. No.: 17/736523

Filed: May 04, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220409398 A1	Dec. 29, 2022

Related U.S. Application Data

continuation-in-part parent-doc US 17665449 20220204 PENDING child-doc US 17736523
continuation-in-part parent-doc US 17515709 20211101 US 12295865 child-doc US 17736523
continuation-in-part parent-doc US 17356950 20210624 US 11612499 child-doc US 17736523

Publication Classification

Int. Cl.: **A61F2/44** (20060101); **A61F2/46** (20060101); A61F2/30 (20060101)

U.S. Cl.:

CPC **A61F2/4611** (20130101); **A61F2/442** (20130101); **A61F2/447** (20130101);
A61F2002/30405 (20130101); A61F2002/30433 (20130101); A61F2002/30556
(20130101); A61F2002/30579 (20130101); A61F2002/4627 (20130101)

Field of Classification Search

CPC: A61F (2/4611); A61F (2/442); A61F (2/447); A61F (2002/30405); A61F (2002/30433);
A61F (2002/30556); A61F (2002/30579); A61F (2002/4627)

USPC: 623/17.11-

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1677337	12/1927	Grove	N/A	N/A
3847154	12/1973	Nordin	N/A	N/A
4553273	12/1984	Wu	N/A	N/A
4636217	12/1986	Ogilvie et al.	N/A	N/A
4716894	12/1987	Lazzeri et al.	N/A	N/A
4759769	12/1987	Hedman et al.	N/A	N/A
5059193	12/1990	Kuslich	N/A	N/A
5171278	12/1991	Pisharodi	N/A	N/A
5228811	12/1992	Potter	N/A	N/A
5284483	12/1993	Johnson et al.	N/A	N/A
5336223	12/1993	Rogers	N/A	N/A
5390683	12/1994	Pisharodi	N/A	N/A
5522899	12/1995	Michelson	N/A	N/A
5554191	12/1995	Lahille et al.	N/A	N/A
5575790	12/1995	Chen et al.	N/A	N/A
5609635	12/1996	Michelson	N/A	N/A
5653762	12/1996	Pisharodi	N/A	N/A
5653763	12/1996	Errico et al.	N/A	N/A
5658336	12/1996	Pisharodi	N/A	N/A
5665122	12/1996	Kambin	N/A	N/A
5693100	12/1996	Pisharodi	N/A	N/A
5697977	12/1996	Pisharodi	N/A	N/A
5702391	12/1996	Lin	N/A	N/A
5702453	12/1996	Rabbe et al.	N/A	N/A

5702455	12/1996	Saggar	N/A	N/A
5797918	12/1997	McGuire et al.	N/A	N/A
5800550	12/1997	Sertich	N/A	N/A
5865848	12/1998	Baker	N/A	N/A
5893890	12/1998	Pisharodi	N/A	N/A
5931777	12/1998	Sava	N/A	N/A
5941885	12/1998	Jackson	N/A	N/A
5971987	12/1998	Huxel et al.	N/A	N/A
5980522	12/1998	Koros et al.	N/A	N/A
6045579	12/1999	Hochshuler et al.	N/A	N/A
6074343	12/1999	Nathanson et al.	N/A	N/A
6080193	12/1999	Hochshuler et al.	N/A	N/A
6099531	12/1999	Bonutti	N/A	N/A
6102949	12/1999	Biedermann et al.	N/A	N/A
6102950	12/1999	Vaccaro	N/A	N/A
6106557	12/1999	Robioneck et al.	N/A	N/A
6113638	12/1999	Williams et al.	N/A	N/A
6117174	12/1999	Nolan	N/A	N/A
6132465	12/1999	Ray et al.	N/A	N/A
6159211	12/1999	Boriani et al.	N/A	N/A
6159244	12/1999	Suddaby	N/A	N/A
6176882	12/2000	Biedermann et al.	N/A	N/A
6179873	12/2000	Zientek	N/A	N/A
6190414	12/2000	Young et al.	N/A	N/A
6193757	12/2000	Foley et al.	N/A	N/A
6217579	12/2000	Koros	N/A	N/A
6245108	12/2000	Biscup	N/A	N/A
6309421	12/2000	Pisharodi	N/A	N/A
6342074	12/2001	Simpson	N/A	N/A
6371989	12/2001	Chauvin et al.	N/A	N/A
6395031	12/2001	Foley et al.	N/A	N/A
6423063	12/2001	Bonutti	N/A	N/A
6432106	12/2001	Fraser	N/A	N/A
6436140	12/2001	Liu et al.	N/A	N/A
6443989	12/2001	Jackson	N/A	N/A
6443990	12/2001	Aebi et al.	N/A	N/A
6454806	12/2001	Cohen et al.	N/A	N/A
6454807	12/2001	Jackson	N/A	N/A
6461359	12/2001	Tribus et al.	N/A	N/A
6475218	12/2001	Gournay et al.	N/A	N/A
6491724	12/2001	Ferree	N/A	N/A
6520991	12/2002	Huene	N/A	N/A
6520993	12/2002	James et al.	N/A	N/A
6524238	12/2002	Velikaris et al.	N/A	N/A
6527803	12/2002	Crozet et al.	N/A	N/A
6562074	12/2002	Gerbec et al.	N/A	N/A
6576016	12/2002	Hochshuler et al.	N/A	N/A
6623525	12/2002	Ralph et al.	N/A	N/A
6629998	12/2002	Lin	N/A	N/A
6635086	12/2002	Lin	N/A	N/A

6648917	12/2002	Gerbec et al.	N/A	N/A
6676703	12/2003	Biscup	N/A	N/A
6685742	12/2003	Jackson	N/A	N/A
6723126	12/2003	Berry	N/A	N/A
6770096	12/2003	Bolger et al.	N/A	N/A
6773460	12/2003	Jackson	N/A	N/A
6821298	12/2003	Jackson	N/A	N/A
6835206	12/2003	Jackson	N/A	N/A
6849093	12/2004	Michelson	N/A	N/A
6852129	12/2004	Gerbec et al.	N/A	N/A
6863673	12/2004	Gerbec et al.	N/A	N/A
6923814	12/2004	Hildebrand et al.	N/A	N/A
6926737	12/2004	Jackson	N/A	N/A
6953477	12/2004	Berry	N/A	N/A
6964687	12/2004	Bernard et al.	N/A	N/A
6974480	12/2004	Messerli et al.	N/A	N/A
6984234	12/2005	Bray	N/A	N/A
7112222	12/2005	Fraser et al.	N/A	N/A
7135043	12/2005	Nakahara et al.	N/A	N/A
7137997	12/2005	Paul	N/A	N/A
7172627	12/2006	Fiere et al.	N/A	N/A
7188626	12/2006	Foley et al.	N/A	N/A
7204853	12/2006	Gordon et al.	N/A	N/A
7232464	12/2006	Mathieu et al.	N/A	N/A
7238203	12/2006	Bagga et al.	N/A	N/A
7255700	12/2006	Kaiser et al.	N/A	N/A
7316532	12/2007	Matthys-Mark	N/A	N/A
7316714	12/2007	Gordon et al.	N/A	N/A
7407483	12/2007	Perez-Cruet et al.	N/A	N/A
7481766	12/2008	Lee et al.	N/A	N/A
7491168	12/2008	Raymond et al.	N/A	N/A
7537565	12/2008	Bass	N/A	N/A
7618456	12/2008	Mathieu et al.	N/A	N/A
7625394	12/2008	Molz, IV et al.	N/A	N/A
7635366	12/2008	Abdou	N/A	N/A
7637909	12/2008	Lechot et al.	N/A	N/A
7655046	12/2009	Dryer et al.	N/A	N/A
7678148	12/2009	Peterman	N/A	N/A
7703727	12/2009	Selness	N/A	N/A
7708778	12/2009	Gordon et al.	N/A	N/A
7708779	12/2009	Edie et al.	N/A	N/A
7727280	12/2009	McLuen	N/A	N/A
7753958	12/2009	Gordon et al.	N/A	N/A
7780594	12/2009	Hutton	N/A	N/A
7806932	12/2009	Webb et al.	N/A	N/A
7815682	12/2009	Peterson et al.	N/A	N/A
7819801	12/2009	Miles et al.	N/A	N/A
7824428	12/2009	Mikkonen et al.	N/A	N/A
7828849	12/2009	Lim	N/A	N/A
7846167	12/2009	Garcia et al.	N/A	N/A

7846207	12/2009	Lechmann et al.	N/A	N/A
7850731	12/2009	Brittan et al.	N/A	N/A
7850733	12/2009	Baynham et al.	N/A	N/A
7862616	12/2010	Lechmann et al.	N/A	N/A
7875076	12/2010	Mathieu et al.	N/A	N/A
7883542	12/2010	Zipnick	N/A	N/A
7892173	12/2010	Miles et al.	N/A	N/A
7909869	12/2010	Gordon et al.	N/A	N/A
7914559	12/2010	Carls et al.	N/A	N/A
7967821	12/2010	Sicvol et al.	N/A	N/A
7981031	12/2010	Frasier et al.	N/A	N/A
8016836	12/2010	Corrao et al.	N/A	N/A
8062375	12/2010	Glerum et al.	N/A	N/A
8105382	12/2011	Olmos et al.	N/A	N/A
8118870	12/2011	Gordon et al.	N/A	N/A
8118871	12/2011	Gordon et al.	N/A	N/A
8123810	12/2011	Gordon et al.	N/A	N/A
8147550	12/2011	Gordon et al.	N/A	N/A
8172903	12/2011	Gordon et al.	N/A	N/A
8182539	12/2011	Tyber et al.	N/A	N/A
8257442	12/2011	Edie et al.	N/A	N/A
8262570	12/2011	White et al.	N/A	N/A
8262662	12/2011	Beardsley et al.	N/A	N/A
8262710	12/2011	Freedman et al.	N/A	N/A
8287597	12/2011	Pimenta et al.	N/A	N/A
8303498	12/2011	Miles et al.	N/A	N/A
8303658	12/2011	Peterman	N/A	N/A
8303663	12/2011	Jimenez et al.	N/A	N/A
8317866	12/2011	Palmatier et al.	N/A	N/A
8323185	12/2011	Perez-Cruet et al.	N/A	N/A
8328872	12/2011	Duffield et al.	N/A	N/A
8343048	12/2012	Warren, Jr.	N/A	N/A
8353826	12/2012	Weiman	N/A	N/A
8355780	12/2012	Miles et al.	N/A	N/A
8382842	12/2012	Greenhalgh et al.	N/A	N/A
8388527	12/2012	Miles et al.	N/A	N/A
8398713	12/2012	Weiman	N/A	N/A
8403990	12/2012	Dryer et al.	N/A	N/A
8419797	12/2012	Biedermann et al.	N/A	N/A
8425528	12/2012	Berry et al.	N/A	N/A
8435298	12/2012	Weiman	N/A	N/A
8480576	12/2012	Sandhu	N/A	N/A
8496706	12/2012	Ragab et al.	N/A	N/A
8500634	12/2012	Miles et al.	N/A	N/A
8500749	12/2012	Lee et al.	N/A	N/A
8506635	12/2012	Palmatier et al.	N/A	N/A
8517935	12/2012	Marchek et al.	N/A	N/A
8518120	12/2012	Glerum et al.	N/A	N/A
8535380	12/2012	Greenhalgh et al.	N/A	N/A
8545566	12/2012	Niemiec et al.	N/A	N/A

8550994	12/2012	Miles et al.	N/A	N/A
8556808	12/2012	Miles et al.	N/A	N/A
8556979	12/2012	Glerum et al.	N/A	N/A
8579809	12/2012	Parker	N/A	N/A
8579898	12/2012	Prandi et al.	N/A	N/A
8579979	12/2012	Edie et al.	N/A	N/A
8579981	12/2012	Lim et al.	N/A	N/A
8602984	12/2012	Raymond et al.	N/A	N/A
8608785	12/2012	Reed et al.	N/A	N/A
8628576	12/2013	Triplett et al.	N/A	N/A
8628578	12/2013	Miller et al.	N/A	N/A
8632595	12/2013	Weiman	N/A	N/A
8641768	12/2013	Duffield et al.	N/A	N/A
8647386	12/2013	Gordon et al.	N/A	N/A
8663329	12/2013	Ernst	N/A	N/A
8668419	12/2013	Hardt et al.	N/A	N/A
8668715	12/2013	Sandhu	N/A	N/A
8679183	12/2013	Glerum et al.	N/A	N/A
8685095	12/2013	Miller et al.	N/A	N/A
8685098	12/2013	Glerum et al.	N/A	N/A
8696559	12/2013	Miles et al.	N/A	N/A
8709083	12/2013	Duffield et al.	N/A	N/A
8709085	12/2013	Lechmann et al.	N/A	N/A
8709086	12/2013	Glerum	N/A	N/A
8715285	12/2013	Lewis et al.	N/A	N/A
8715353	12/2013	Bagga et al.	N/A	N/A
8740983	12/2013	Arnold	623/17.16	A61F 2/4611
8753271	12/2013	Miles et al.	N/A	N/A
8753396	12/2013	Hockett et al.	N/A	N/A
8764649	12/2013	Miles et al.	N/A	N/A
8771360	12/2013	Jimenez et al.	N/A	N/A
8778025	12/2013	Ragab et al.	N/A	N/A
8778027	12/2013	Medina	N/A	N/A
8795366	12/2013	Varela	N/A	N/A
8808304	12/2013	Weiman et al.	N/A	N/A
8808305	12/2013	Kleiner	N/A	N/A
8827902	12/2013	Dietze, Jr. et al.	N/A	N/A
8828085	12/2013	Jensen	N/A	N/A
8840668	12/2013	Donahoe et al.	N/A	N/A
8845731	12/2013	Weiman	N/A	N/A
8845732	12/2013	Weiman	N/A	N/A
8845734	12/2013	Weiman	N/A	N/A
8852252	12/2013	Venturini et al.	N/A	N/A
8852282	12/2013	Farley et al.	N/A	N/A
8864833	12/2013	Glerum et al.	N/A	N/A
8882813	12/2013	Jones et al.	N/A	N/A
8888853	12/2013	Glerum et al.	N/A	N/A
8894708	12/2013	Thalgott et al.	N/A	N/A
8894711	12/2013	Varela	N/A	N/A
8894712	12/2013	Varela	N/A	N/A

8906095	12/2013	Christensen et al.	N/A	N/A
8920500	12/2013	Pimenta et al.	N/A	N/A
8926704	12/2014	Glerum et al.	N/A	N/A
8936641	12/2014	Cain	N/A	N/A
8940049	12/2014	Jimenez et al.	N/A	N/A
8968363	12/2014	Weiman et al.	N/A	N/A
8986344	12/2014	Sandhu	N/A	N/A
8992425	12/2014	Karpowicz et al.	N/A	N/A
8992544	12/2014	Sasing	N/A	N/A
8998906	12/2014	Kirschman	N/A	N/A
9005292	12/2014	Melamed	N/A	N/A
9005293	12/2014	Moskowitz et al.	N/A	N/A
9005295	12/2014	Kueenzi et al.	N/A	N/A
9017412	12/2014	Wolters et al.	N/A	N/A
9034045	12/2014	Davenport et al.	N/A	N/A
9050146	12/2014	Woolley et al.	N/A	N/A
9050194	12/2014	Thibodeau	N/A	N/A
9060877	12/2014	Kleiner	N/A	N/A
9072548	12/2014	Matityahu	N/A	N/A
9072563	12/2014	Garcia et al.	N/A	N/A
9084591	12/2014	Reglos et al.	N/A	N/A
9113854	12/2014	Ellman	N/A	N/A
9119730	12/2014	Glerum et al.	N/A	N/A
9125757	12/2014	Weiman	N/A	N/A
9132021	12/2014	Mermuys et al.	N/A	N/A
9138217	12/2014	Smith et al.	N/A	N/A
9138330	12/2014	Hansell et al.	N/A	N/A
9138331	12/2014	Aferzon	N/A	N/A
9149367	12/2014	Davenport et al.	N/A	N/A
9155628	12/2014	Glerum et al.	N/A	N/A
9155631	12/2014	Seifert et al.	N/A	N/A
9161841	12/2014	Kana et al.	N/A	N/A
9179903	12/2014	Cianfrani et al.	N/A	N/A
9179952	12/2014	Biedermann et al.	N/A	N/A
9186193	12/2014	Kleiner et al.	N/A	N/A
9186258	12/2014	Davenport et al.	N/A	N/A
9192482	12/2014	Pimenta et al.	N/A	N/A
9192483	12/2014	Radcliffe et al.	N/A	N/A
9198772	12/2014	Weiman	N/A	N/A
9204972	12/2014	Weiman et al.	N/A	N/A
9204974	12/2014	Glerum et al.	N/A	N/A
9211194	12/2014	Bagga et al.	N/A	N/A
9211196	12/2014	Glerum et al.	N/A	N/A
9216095	12/2014	Glerum et al.	N/A	N/A
9226836	12/2015	Glerum	N/A	N/A
9233007	12/2015	Sungarian et al.	N/A	N/A
9233009	12/2015	Gray et al.	N/A	N/A
9233010	12/2015	Thalgott et al.	N/A	N/A
9259327	12/2015	Niemiec et al.	N/A	N/A
9271846	12/2015	Lim et al.	N/A	N/A

9308099	12/2015	Triplett et al.	N/A	N/A
9320610	12/2015	Alheidt et al.	N/A	N/A
9351845	12/2015	Pimenta et al.	N/A	N/A
9351848	12/2015	Glerum et al.	N/A	N/A
9357909	12/2015	Perez-Cruet et al.	N/A	N/A
9358126	12/2015	Glerum et al.	N/A	N/A
9358127	12/2015	Duffield et al.	N/A	N/A
9358128	12/2015	Glerum et al.	N/A	N/A
9358129	12/2015	Weiman	N/A	N/A
9364341	12/2015	Gowan	N/A	N/A
9364343	12/2015	Duffield et al.	N/A	N/A
9370434	12/2015	Weiman	N/A	N/A
9370435	12/2015	Walkenhorst et al.	N/A	N/A
9381008	12/2015	Thornburg	N/A	N/A
9386916	12/2015	Predick et al.	N/A	N/A
9387092	12/2015	Mermuys et al.	N/A	N/A
9402673	12/2015	Cormier et al.	N/A	N/A
9402739	12/2015	Weiman et al.	N/A	N/A
9408596	12/2015	Blain	N/A	N/A
9408708	12/2015	Greenhalgh	N/A	N/A
9414828	12/2015	Abidin et al.	N/A	N/A
9414934	12/2015	Cain	N/A	N/A
9414937	12/2015	Carlson et al.	N/A	N/A
9421110	12/2015	Masson et al.	N/A	N/A
9427331	12/2015	Amin	N/A	N/A
9445919	12/2015	Palmatier et al.	N/A	N/A
9452063	12/2015	Glerum et al.	N/A	N/A
9456903	12/2015	Glerum et al.	N/A	N/A
9456906	12/2015	Gray et al.	N/A	N/A
9468405	12/2015	Miles et al.	N/A	N/A
9474622	12/2015	McLaughlin et al.	N/A	N/A
9474625	12/2015	Weiman	N/A	N/A
9480573	12/2015	Perloff et al.	N/A	N/A
9480576	12/2015	Pepper et al.	N/A	N/A
9480579	12/2015	Davenport et al.	N/A	N/A
9486133	12/2015	Lee et al.	N/A	N/A
9486325	12/2015	Davenport et al.	N/A	N/A
9486327	12/2015	Martynova et al.	N/A	N/A
9486328	12/2015	Jimenez et al.	N/A	N/A
9492287	12/2015	Glerum et al.	N/A	N/A
9492288	12/2015	Wagner et al.	N/A	N/A
9492289	12/2015	Davenport et al.	N/A	N/A
9498349	12/2015	Patterson et al.	N/A	N/A
9510954	12/2015	Glerum et al.	N/A	N/A
9517098	12/2015	Anderson	N/A	N/A
9522070	12/2015	Flower et al.	N/A	N/A
9526620	12/2015	Slivka et al.	N/A	N/A
9526625	12/2015	Cain	N/A	N/A
9532821	12/2016	Moskowitz et al.	N/A	N/A
9539103	12/2016	McLaughlin et al.	N/A	N/A

9539108	12/2016	Glerum et al.	N/A	N/A
9545320	12/2016	Padovani et al.	N/A	N/A
9549723	12/2016	Hynes et al.	N/A	N/A
9549824	12/2016	McAfee	N/A	N/A
9561116	12/2016	Weiman et al.	N/A	N/A
9566163	12/2016	Suddaby et al.	N/A	N/A
9566166	12/2016	Parry et al.	N/A	N/A
9566168	12/2016	Glerum et al.	N/A	N/A
9572560	12/2016	Mast et al.	N/A	N/A
9572677	12/2016	Davenport et al.	N/A	N/A
9572681	12/2016	Mathieu et al.	N/A	N/A
9579124	12/2016	Gordon et al.	N/A	N/A
9579139	12/2016	Cormier et al.	N/A	N/A
9579213	12/2016	Bal et al.	N/A	N/A
9585649	12/2016	Blain et al.	N/A	N/A
9585762	12/2016	Suddaby et al.	N/A	N/A
9585766	12/2016	Robinson	N/A	N/A
9585767	12/2016	Robinson	N/A	N/A
9592129	12/2016	Slivka et al.	N/A	N/A
9597195	12/2016	Cain	N/A	N/A
9603643	12/2016	Reed et al.	N/A	N/A
9603713	12/2016	Moskowitz et al.	N/A	N/A
9603717	12/2016	Barra et al.	N/A	N/A
9615818	12/2016	Baudouin et al.	N/A	N/A
9615936	12/2016	Duffield et al.	N/A	N/A
9622732	12/2016	Martinelli et al.	N/A	N/A
9622875	12/2016	Moskowitz et al.	N/A	N/A
9622876	12/2016	Greenhalgh et al.	N/A	N/A
9629729	12/2016	Grimberg, Jr. et al.	N/A	N/A
9636097	12/2016	Bass	N/A	N/A
9642720	12/2016	Radcliffe et al.	N/A	N/A
9649198	12/2016	Wolters et al.	N/A	N/A
9655746	12/2016	Seifert	N/A	N/A
9655747	12/2016	Glerum et al.	N/A	N/A
9662224	12/2016	Weiman et al.	N/A	N/A
9668784	12/2016	Brumfield et al.	N/A	N/A
9668876	12/2016	Blain et al.	N/A	N/A
9668879	12/2016	Jimenez et al.	N/A	N/A
9675465	12/2016	Padovani et al.	N/A	N/A
9675467	12/2016	Duffield et al.	N/A	N/A
9675468	12/2016	Jensen	N/A	N/A
9693871	12/2016	Richerme et al.	N/A	N/A
9700428	12/2016	Niemiec et al.	N/A	N/A
9707092	12/2016	Davenport et al.	N/A	N/A
9713536	12/2016	Foley et al.	N/A	N/A
9717601	12/2016	Miller	N/A	N/A
9730684	12/2016	Beale et al.	N/A	N/A
9730806	12/2016	Capote	N/A	N/A
9737288	12/2016	Karpowicz et al.	N/A	N/A
9750617	12/2016	Lim et al.	N/A	N/A

9750618	12/2016	Daffinon et al.	N/A	N/A
9757249	12/2016	Radcliffe et al.	N/A	N/A
9763722	12/2016	Roybal	N/A	N/A
9770343	12/2016	Weiman	N/A	N/A
9782265	12/2016	Weiman et al.	N/A	N/A
9788971	12/2016	Stein	N/A	N/A
9795370	12/2016	O'Connell et al.	N/A	N/A
9795371	12/2016	Miles et al.	N/A	N/A
9801733	12/2016	Wolters et al.	N/A	N/A
9801734	12/2016	Stein et al.	N/A	N/A
9808352	12/2016	Suddaby et al.	N/A	N/A
9826966	12/2016	Mast et al.	N/A	N/A
9827024	12/2016	Cormier et al.	N/A	N/A
9827107	12/2016	Arnin	N/A	N/A
9833333	12/2016	Duffield et al.	N/A	N/A
9833336	12/2016	Davenport et al.	N/A	N/A
9839527	12/2016	Robinson	N/A	N/A
9839528	12/2016	Weiman et al.	N/A	N/A
9848993	12/2016	Moskowitz et al.	N/A	N/A
9848996	12/2016	Faulhaber	N/A	N/A
9855151	12/2017	Weiman	N/A	N/A
9867715	12/2017	McLaughlin et al.	N/A	N/A
9872779	12/2017	Miller et al.	N/A	N/A
9889019	12/2017	Rogers et al.	N/A	N/A
9907671	12/2017	Fessler	N/A	N/A
9907673	12/2017	Weiman et al.	N/A	N/A
9918709	12/2017	Sandhu	N/A	N/A
9924859	12/2017	Lee et al.	N/A	N/A
9924940	12/2017	Moskowitz et al.	N/A	N/A
9925062	12/2017	Glerum et al.	N/A	N/A
9925064	12/2017	Duffield et al.	N/A	N/A
9931223	12/2017	Cain	N/A	N/A
9937053	12/2017	Melkent et al.	N/A	N/A
9937060	12/2017	Fuhrer	N/A	A61F 2/442
9943342	12/2017	Tanaka et al.	N/A	N/A
9943418	12/2017	Davenport et al.	N/A	N/A
9949775	12/2017	Reed et al.	N/A	N/A
9949841	12/2017	Glerum et al.	N/A	N/A
9956087	12/2017	Seifert et al.	N/A	N/A
9962202	12/2017	Anderson	N/A	N/A
9962270	12/2017	Alheidt et al.	N/A	N/A
9962271	12/2017	Glerum	N/A	N/A
9962272	12/2017	Daffinon et al.	N/A	N/A
9968461	12/2017	Zappacosta et al.	N/A	N/A
9968462	12/2017	Weiman	N/A	N/A
9974531	12/2017	Miles et al.	N/A	N/A
9974662	12/2017	Hessler et al.	N/A	N/A
9974664	12/2017	Emerick et al.	N/A	N/A
9980825	12/2017	Nichols et al.	N/A	N/A
9980826	12/2017	Martynova et al.	N/A	N/A

9987141	12/2017	Duffield et al.	N/A	N/A
9987143	12/2017	Robinson et al.	N/A	N/A
9987144	12/2017	Seifert et al.	N/A	N/A
9987146	12/2017	Lentner et al.	N/A	N/A
9993239	12/2017	Karpowicz et al.	N/A	N/A
9993350	12/2017	Cain	N/A	N/A
10004607	12/2017	Weiman et al.	N/A	N/A
10004608	12/2017	Carnes et al.	N/A	N/A
10016282	12/2017	Seifert et al.	N/A	N/A
10016284	12/2017	Moskowitz et al.	N/A	N/A
10022239	12/2017	Entner et al.	N/A	N/A
10028842	12/2017	Gray et al.	N/A	N/A
10034765	12/2017	Blain et al.	N/A	N/A
10034769	12/2017	Baynham	N/A	N/A
10034771	12/2017	Capote et al.	N/A	N/A
10034772	12/2017	Glerum et al.	N/A	N/A
10034773	12/2017	McLaughlin et al.	N/A	N/A
10039539	12/2017	Friedrich et al.	N/A	N/A
10039650	12/2017	Lamborne et al.	N/A	N/A
10052214	12/2017	Jimenez et al.	N/A	N/A
10058431	12/2017	Tyber et al.	N/A	N/A
10060469	12/2017	Jimenez et al.	N/A	N/A
10070852	12/2017	Mast et al.	N/A	N/A
10076320	12/2017	Mast et al.	N/A	N/A
10076423	12/2017	Miller et al.	N/A	N/A
10080666	12/2017	Suddaby et al.	N/A	N/A
10080669	12/2017	Davenport et al.	N/A	N/A
10085846	12/2017	Grotz	N/A	N/A
10085849	12/2017	Weiman et al.	N/A	N/A
10092417	12/2017	Weiman et al.	N/A	N/A
10098758	12/2017	Matthews et al.	N/A	N/A
10098759	12/2017	Weiman	N/A	N/A
10111755	12/2017	Foley et al.	N/A	N/A
10111758	12/2017	Robinson	N/A	N/A
10117754	12/2017	Davenport et al.	N/A	N/A
10117755	12/2017	Emerick et al.	N/A	N/A
10137002	12/2017	Padovani et al.	N/A	N/A
10137006	12/2017	Dewey et al.	N/A	N/A
10137007	12/2017	Dewey et al.	N/A	N/A
10137009	12/2017	Weiman et al.	N/A	N/A
10149671	12/2017	Predick et al.	N/A	N/A
10149710	12/2017	Tanaka et al.	N/A	N/A
10154781	12/2017	Weiman	N/A	N/A
10154912	12/2017	Glerum	N/A	N/A
10154914	12/2017	Robinson	N/A	N/A
10159584	12/2017	Carnes et al.	N/A	N/A
10166117	12/2018	Daffinson et al.	N/A	N/A
10172515	12/2018	Lee et al.	N/A	N/A
10172652	12/2018	Woolley et al.	N/A	N/A
10178987	12/2018	Predick et al.	N/A	N/A

10179053	12/2018	Zappacosta et al.	N/A	N/A
10182922	12/2018	Nichols et al.	N/A	N/A
10188527	12/2018	Rogers et al.	N/A	N/A
10195050	12/2018	Palmatier et al.	N/A	N/A
10201431	12/2018	Slater et al.	N/A	N/A
10213192	12/2018	Capote	N/A	N/A
10213193	12/2018	Karpowicz et al.	N/A	N/A
10219798	12/2018	Capote	N/A	N/A
10219913	12/2018	Matthews et al.	N/A	N/A
10219914	12/2018	Faulhaber	N/A	N/A
10219915	12/2018	Stein	N/A	N/A
10226356	12/2018	Grotz	N/A	N/A
10226359	12/2018	Glerum et al.	N/A	N/A
10238375	12/2018	O'Connell et al.	N/A	N/A
10238383	12/2018	Moskowitz et al.	N/A	N/A
10238503	12/2018	Branch et al.	N/A	N/A
10245015	12/2018	Predick et al.	N/A	N/A
10251643	12/2018	Moskowitz et al.	N/A	N/A
10265191	12/2018	Lim et al.	N/A	N/A
10278686	12/2018	Baudouin et al.	N/A	N/A
10278786	12/2018	Friedrich et al.	N/A	N/A
10278830	12/2018	Walker et al.	N/A	N/A
10278831	12/2018	Sandul	N/A	N/A
10278832	12/2018	Nichols et al.	N/A	N/A
10285680	12/2018	Friedrich et al.	N/A	N/A
10285819	12/2018	Greenhalgh	N/A	N/A
10285824	12/2018	Robinson	N/A	N/A
10292828	12/2018	Greenhalgh	N/A	N/A
10299777	12/2018	Mast et al.	N/A	N/A
10299934	12/2018	Seifert et al.	N/A	N/A
10299937	12/2018	McAfee	N/A	N/A
10307268	12/2018	Moskowitz et al.	N/A	N/A
10314622	12/2018	Brumfield et al.	N/A	N/A
10314719	12/2018	Hessler et al.	N/A	N/A
10322007	12/2018	Masson et al.	N/A	N/A
10322009	12/2018	Aghayev et al.	N/A	N/A
10327909	12/2018	Baynham	N/A	N/A
10327912	12/2018	Suddaby	N/A	N/A
10327917	12/2018	Glerum et al.	N/A	N/A
10342675	12/2018	Alheidt	N/A	N/A
10350085	12/2018	Glerum et al.	N/A	N/A
10357233	12/2018	Miles et al.	N/A	N/A
10363142	12/2018	McClintock et al.	N/A	N/A
10363144	12/2018	Overes et al.	N/A	N/A
10369004	12/2018	Faulhaber	N/A	N/A
10369008	12/2018	Jimenez et al.	N/A	N/A
10369010	12/2018	Robinson et al.	N/A	N/A
10369012	12/2018	Fessler	N/A	N/A
10376377	12/2018	Seifert et al.	N/A	N/A
10390962	12/2018	Weiman	N/A	N/A

10390964	12/2018	Faulhaber	N/A	N/A
10398563	12/2018	Engstrom	N/A	N/A
10398566	12/2018	Olmos et al.	N/A	N/A
10413419	12/2018	Thibodeau	N/A	N/A
10413422	12/2018	Flower et al.	N/A	N/A
10413423	12/2018	Overes et al.	N/A	N/A
10426450	12/2018	Vogel et al.	N/A	N/A
10426633	12/2018	Moskowitz et al.	N/A	N/A
10426634	12/2018	Al-Jazaeri et al.	N/A	N/A
10441430	12/2018	Ludwig et al.	N/A	N/A
10449056	12/2018	Cain	N/A	N/A
10456122	12/2018	Koltz et al.	N/A	N/A
10470894	12/2018	Foley et al.	N/A	N/A
10478319	12/2018	Moskowitz et al.	N/A	N/A
10492912	12/2018	Gregersen et al.	N/A	N/A
10492922	12/2018	Mathieu et al.	N/A	N/A
10492924	12/2018	Stein et al.	N/A	N/A
10500064	12/2018	Robinson	N/A	N/A
10512550	12/2018	Bechtel et al.	N/A	N/A
10517645	12/2018	van der Pol	N/A	N/A
10524924	12/2019	Davenport et al.	N/A	N/A
10531903	12/2019	Daly et al.	N/A	N/A
10537436	12/2019	Maguire et al.	N/A	N/A
10537438	12/2019	Martynova et al.	N/A	N/A
10555729	12/2019	Cole et al.	N/A	N/A
10561411	12/2019	Cole et al.	N/A	N/A
10575889	12/2019	Roybal	N/A	N/A
10575960	12/2019	Duffield et al.	N/A	N/A
10582959	12/2019	Langer et al.	N/A	N/A
10583015	12/2019	Olmos et al.	N/A	N/A
10603078	12/2019	Simpson et al.	N/A	N/A
10610376	12/2019	Kuyler et al.	N/A	N/A
10624757	12/2019	Bost et al.	N/A	N/A
10624758	12/2019	Slivka et al.	N/A	N/A
10624761	12/2019	Davenport et al.	N/A	N/A
10639163	12/2019	Tyber et al.	N/A	N/A
10639166	12/2019	Weiman et al.	N/A	N/A
10653458	12/2019	Tanaka et al.	N/A	N/A
10667925	12/2019	Emerick et al.	N/A	N/A
10667927	12/2019	Lamborne et al.	N/A	N/A
10675157	12/2019	Zakelj et al.	N/A	N/A
10682241	12/2019	Glerum et al.	N/A	N/A
10687963	12/2019	Jimenez et al.	N/A	N/A
10702393	12/2019	Davenport et al.	N/A	N/A
10709569	12/2019	McLaughlin et al.	N/A	N/A
10709571	12/2019	Iott et al.	N/A	N/A
10709572	12/2019	Daffinson et al.	N/A	N/A
10709575	12/2019	Robinson	N/A	N/A
10722377	12/2019	Glerum et al.	N/A	N/A
10722379	12/2019	McLaughlin et al.	N/A	N/A

10729561	12/2019	Glerum	N/A	N/A
10743858	12/2019	Cole et al.	N/A	N/A
10744002	12/2019	Glerum et al.	N/A	N/A
10758366	12/2019	Daffinson et al.	N/A	N/A
10758367	12/2019	Weiman et al.	N/A	N/A
10758369	12/2019	Rogers et al.	N/A	N/A
10765528	12/2019	Weiman et al.	N/A	N/A
10772737	12/2019	Gray et al.	N/A	N/A
10779955	12/2019	Kuyler et al.	N/A	N/A
10779957	12/2019	Neiman et al.	N/A	N/A
10786364	12/2019	Davenport et al.	N/A	N/A
10786369	12/2019	Carnes et al.	N/A	N/A
10799368	12/2019	Glerum et al.	N/A	N/A
10835387	12/2019	Weiman et al.	N/A	N/A
10842640	12/2019	Weiman et al.	N/A	N/A
10842644	12/2019	Weiman et al.	N/A	N/A
10856997	12/2019	Cowan et al.	N/A	N/A
10869769	12/2019	Eisen et al.	N/A	N/A
10874447	12/2019	Tanaka et al.	N/A	N/A
10874522	12/2019	Weiman	N/A	N/A
10874523	12/2019	Weiman et al.	N/A	N/A
10874524	12/2019	Bjork	N/A	N/A
10881524	12/2020	Eisen et al.	N/A	N/A
10881531	12/2020	Berry	N/A	N/A
10888431	12/2020	Robinson	N/A	N/A
10898344	12/2020	Alheidt et al.	N/A	N/A
10898346	12/2020	Suddaby	N/A	N/A
10925656	12/2020	Cole et al.	N/A	N/A
10925750	12/2020	Zappacosta et al.	N/A	N/A
10925752	12/2020	Weiman	N/A	N/A
10932920	12/2020	Dewey et al.	N/A	N/A
10940014	12/2020	Greenhalgh	N/A	N/A
10945858	12/2020	Bechtel et al.	N/A	N/A
10952866	12/2020	Warren et al.	N/A	N/A
10959855	12/2020	Miller et al.	N/A	N/A
10959856	12/2020	Seifert et al.	N/A	N/A
10973649	12/2020	Weiman et al.	N/A	N/A
10973650	12/2020	Stein	N/A	N/A
10980642	12/2020	Glerum et al.	N/A	N/A
10980644	12/2020	Purcell et al.	N/A	N/A
10993814	12/2020	Wolters	N/A	N/A
11007067	12/2020	Masson et al.	N/A	N/A
11013617	12/2020	Weiman et al.	N/A	N/A
11020238	12/2020	Nichols et al.	N/A	N/A
11020239	12/2020	Miller et al.	N/A	N/A
11026804	12/2020	Jimenez et al.	N/A	N/A
11026812	12/2020	Daffinson et al.	N/A	N/A
11033401	12/2020	Shoshtaev	N/A	N/A
11033402	12/2020	Melkent et al.	N/A	N/A
11033404	12/2020	Faulhaber	N/A	N/A

11039935	12/2020	McAfee	N/A	N/A
11045326	12/2020	Seifert et al.	N/A	N/A
11045327	12/2020	Nichols et al.	N/A	N/A
11051949	12/2020	Walker et al.	N/A	N/A
11051951	12/2020	Robinson et al.	N/A	N/A
11058469	12/2020	Mahajan et al.	N/A	N/A
11065127	12/2020	Entner et al.	N/A	N/A
11065129	12/2020	Sandul	N/A	N/A
11065130	12/2020	Branch et al.	N/A	N/A
11076966	12/2020	Faulhaber	N/A	N/A
11083584	12/2020	Lauf et al.	N/A	N/A
11083595	12/2020	Robinson	N/A	N/A
11090167	12/2020	Emerick et al.	N/A	N/A
11096795	12/2020	Padovani et al.	N/A	N/A
11096797	12/2020	Moskowitz et al.	N/A	N/A
11103366	12/2020	Glerum et al.	N/A	N/A
RE48719	12/2020	Suddaby et al.	N/A	N/A
11109980	12/2020	Seifert et al.	N/A	N/A
11116644	12/2020	Marrocco et al.	N/A	N/A
11123198	12/2020	Black et al.	N/A	N/A
11123200	12/2020	Faulhaber	N/A	N/A
11129731	12/2020	Miller et al.	N/A	N/A
11135071	12/2020	Dewey et al.	N/A	N/A
11147680	12/2020	Tyber et al.	N/A	N/A
11154404	12/2020	Freedman et al.	N/A	N/A
11160666	12/2020	Burkhardt et al.	N/A	N/A
11160669	12/2020	Rogers et al.	N/A	N/A
11166826	12/2020	Huang	N/A	N/A
11173044	12/2020	Jones et al.	N/A	N/A
11179234	12/2020	Dacosta et al.	N/A	N/A
11285014	12/2021	Josse et al.	N/A	N/A
11376134	12/2021	Dewey et al.	N/A	N/A
11617658	12/2022	Josse et al.	N/A	N/A
11723780	12/2022	Seifert et al.	N/A	N/A
11737892	12/2022	Kadaba et al.	N/A	N/A
12064354	12/2023	Robinson et al.	N/A	N/A
2002/0045943	12/2001	Uk	N/A	N/A
2002/0045945	12/2001	Liu et al.	N/A	N/A
2002/0055741	12/2001	Schlapfer et al.	N/A	N/A
2002/0116066	12/2001	Chauvin et al.	N/A	N/A
2002/0128713	12/2001	Ferree	N/A	N/A
2002/0151976	12/2001	Foley et al.	N/A	N/A
2002/0183762	12/2001	Anderson et al.	N/A	N/A
2003/0050701	12/2002	Michelson	N/A	N/A
2003/0130739	12/2002	Gerbec et al.	N/A	N/A
2003/0163132	12/2002	Chin	N/A	N/A
2004/0102778	12/2003	Huebner et al.	N/A	N/A
2004/0172134	12/2003	Berry	N/A	N/A
2004/0186570	12/2003	Rapp	N/A	N/A
2004/0193158	12/2003	Lim et al.	N/A	N/A

2004/0204713	12/2003	Abdou	N/A	N/A
2004/0249461	12/2003	Ferree	N/A	N/A
2004/0254643	12/2003	Jackson	N/A	N/A
2004/0254644	12/2003	Taylor	N/A	N/A
2005/0015094	12/2004	Keller	N/A	N/A
2005/0015149	12/2004	Michelson	N/A	N/A
2005/0033429	12/2004	Kuo	N/A	N/A
2005/0033439	12/2004	Gordon et al.	N/A	N/A
2005/0147478	12/2004	Greenberg	N/A	N/A
2005/0154459	12/2004	Wolek et al.	N/A	N/A
2005/0209698	12/2004	Gordon et al.	N/A	N/A
2005/0228398	12/2004	Rathbun et al.	N/A	N/A
2006/0122701	12/2005	Kiester	N/A	N/A
2006/0129244	12/2005	Ensign	N/A	N/A
2006/0260446	12/2005	Chang	N/A	N/A
2006/0276901	12/2005	Zipnick et al.	N/A	N/A
2007/0173840	12/2006	Huebner	N/A	N/A
2007/0218750	12/2006	Corrao et al.	N/A	N/A
2007/0233150	12/2006	Blain et al.	N/A	N/A
2007/0270859	12/2006	Companiononi et al.	N/A	N/A
2008/0058804	12/2007	Lechot et al.	N/A	N/A
2008/0132959	12/2007	Mikkonen et al.	N/A	N/A
2008/0140207	12/2007	Olmos et al.	N/A	N/A
2008/0287957	12/2007	Hester et al.	N/A	N/A
2009/0024158	12/2008	Viker	N/A	N/A
2009/0036746	12/2008	Blackwell et al.	N/A	N/A
2009/0093830	12/2008	Miller	N/A	N/A
2009/0105834	12/2008	Hovda et al.	N/A	N/A
2009/0292361	12/2008	Lopez	N/A	N/A
2010/0070035	12/2009	Mayer	N/A	N/A
2010/0076440	12/2009	Pamichev et al.	N/A	N/A
2010/0082109	12/2009	Greenhalgh et al.	N/A	N/A
2010/0114183	12/2009	Wassinger et al.	N/A	N/A
2010/0152853	12/2009	Kirschman	N/A	N/A
2010/0191336	12/2009	Greenhalgh	N/A	N/A
2010/0211176	12/2009	Greenhalgh	N/A	N/A
2010/0280617	12/2009	Coppes et al.	N/A	N/A
2010/0286777	12/2009	Errico et al.	N/A	N/A
2011/0118843	12/2010	Mathieu et al.	N/A	N/A
2011/0130838	12/2010	Morgenstern Lopez	N/A	N/A
2011/0144755	12/2010	Baynham et al.	N/A	N/A
2011/0153020	12/2010	Abdelgany et al.	N/A	N/A
2011/0218572	12/2010	Lechmann et al.	N/A	N/A
2011/0218631	12/2010	Woodburn, Sr. et al.	N/A	N/A
2011/0237898	12/2010	Stone et al.	N/A	N/A
2011/0301577	12/2010	Simmen et al.	N/A	N/A
2012/0004732	12/2011	Goel et al.	N/A	N/A
2012/0010717	12/2011	Spann	N/A	N/A
2012/0095515	12/2011	Hamilton	N/A	N/A
2012/0101581	12/2011	Mathieu et al.	N/A	N/A

2012/0109142	12/2011	Dayan	N/A	N/A
2012/0109309	12/2011	Mathieu et al.	N/A	N/A
2012/0109310	12/2011	Mathieu et al.	N/A	N/A
2012/0109312	12/2011	Mathieu et al.	N/A	N/A
2012/0109313	12/2011	Mathieu et al.	N/A	N/A
2012/0123546	12/2011	Medina	N/A	N/A
2012/0143195	12/2011	Sander	N/A	N/A
2012/0150237	12/2011	Combrowski	N/A	N/A
2012/0197401	12/2011	Duncan et al.	N/A	N/A
2012/0209385	12/2011	Aferzon	N/A	N/A
2012/0215313	12/2011	Saidha et al.	N/A	N/A
2012/0215316	12/2011	Mohr et al.	N/A	N/A
2012/0232349	12/2011	Perrow	N/A	N/A
2013/0103095	12/2012	Brumfield	606/301	A61B 17/7064
2013/0158664	12/2012	Palmatier et al.	N/A	N/A
2013/0184823	12/2012	Malberg	N/A	N/A
2013/0190876	12/2012	Drochner et al.	N/A	N/A
2013/0211526	12/2012	Alheidt et al.	N/A	N/A
2013/0226191	12/2012	Thoren et al.	N/A	N/A
2013/0231747	12/2012	Olmos et al.	N/A	N/A
2013/0274557	12/2012	Bowman et al.	N/A	N/A
2013/0304136	12/2012	Gourlaouen- Preissler et al.	N/A	N/A
2013/0317312	12/2012	Eastlack et al.	N/A	N/A
2014/0018816	12/2013	Fenn et al.	N/A	N/A
2014/0039622	12/2013	Glerum et al.	N/A	N/A
2014/0107790	12/2013	Combrowski	N/A	N/A
2014/0114321	12/2013	Davenport et al.	N/A	N/A
2014/0114420	12/2013	Robinson	N/A	N/A
2014/0148904	12/2013	Robinson	N/A	N/A
2014/0163682	12/2013	Iott et al.	N/A	N/A
2014/0180419	12/2013	Dmuschewsky	N/A	N/A
2014/0194992	12/2013	Medina	N/A	N/A
2014/0249629	12/2013	Moskowitz et al.	N/A	N/A
2014/0249631	12/2013	Weiman	N/A	N/A
2014/0277471	12/2013	Gray et al.	N/A	N/A
2014/0277473	12/2013	Perrow	N/A	N/A
2014/0277487	12/2013	Davenport et al.	N/A	N/A
2014/0277500	12/2013	Logan et al.	N/A	N/A
2014/0303674	12/2013	Sasing	N/A	N/A
2014/0364855	12/2013	Stoll et al.	N/A	N/A
2014/0379085	12/2013	Duffield et al.	N/A	N/A
2015/0173915	12/2014	Laubert et al.	N/A	N/A
2015/0223945	12/2014	Weiman et al.	N/A	N/A
2015/0230931	12/2014	Greenhalgh	N/A	N/A
2015/0238236	12/2014	Sasing	N/A	N/A
2015/0354635	12/2014	McClymont et al.	N/A	N/A
2015/0374507	12/2014	Wolters et al.	N/A	N/A
2016/0008924	12/2015	Canourgues et al.	N/A	N/A

2016/0022434	12/2015	Robinson	N/A	N/A
2016/0051373	12/2015	Faulhaber	N/A	N/A
2016/0058571	12/2015	McLaughlin et al.	N/A	N/A
2016/0081681	12/2015	Waugh et al.	N/A	N/A
2016/0089247	12/2015	Nichols et al.	N/A	N/A
2016/0095710	12/2015	Juszczyk et al.	N/A	N/A
2016/0095718	12/2015	Burkhardt et al.	N/A	N/A
2016/0100951	12/2015	Suddaby	623/17.16	A61F 2/442
2016/0128847	12/2015	Kurtaliaj et al.	N/A	N/A
2016/0199073	12/2015	Nino et al.	N/A	N/A
2016/0242930	12/2015	Duffield et al.	N/A	N/A
2016/0256291	12/2015	Miller	N/A	N/A
2016/0278830	12/2015	Arrington	N/A	N/A
2016/0296340	12/2015	Gordon et al.	N/A	N/A
2016/0310291	12/2015	Greenhalgh	N/A	N/A
2016/0345952	12/2015	Kucharzyk et al.	N/A	N/A
2016/0367377	12/2015	Faulhaber	N/A	N/A
2017/0010025	12/2016	Mayershofer	N/A	N/A
2017/0029635	12/2016	Doll et al.	N/A	N/A
2017/0035406	12/2016	Abidin et al.	N/A	N/A
2017/0049651	12/2016	Lim et al.	N/A	N/A
2017/0049653	12/2016	Lim et al.	N/A	N/A
2017/0095345	12/2016	Davenport et al.	N/A	N/A
2017/0100255	12/2016	Hleihil et al.	N/A	N/A
2017/0100257	12/2016	Weiman et al.	N/A	N/A
2017/0105844	12/2016	Kuyler et al.	N/A	N/A
2017/0112630	12/2016	Kuyler et al.	N/A	N/A
2017/0151065	12/2016	Warren et al.	N/A	N/A
2017/0156882	12/2016	Rathbun et al.	N/A	N/A
2017/0156884	12/2016	Rathbun et al.	N/A	N/A
2017/0189200	12/2016	Miller et al.	N/A	N/A
2017/0189204	12/2016	Riemhofer et al.	N/A	N/A
2017/0202678	12/2016	Duffield et al.	N/A	N/A
2017/0215856	12/2016	Martinelli et al.	N/A	N/A
2017/0224502	12/2016	Wolters et al.	N/A	N/A
2017/0224504	12/2016	Butler et al.	N/A	N/A
2017/0231675	12/2016	Combrowski	N/A	N/A
2017/0231780	12/2016	D'Urso	N/A	N/A
2017/0246006	12/2016	Carnes et al.	N/A	N/A
2017/0290677	12/2016	Olmos et al.	N/A	N/A
2017/0296352	12/2016	Richerme et al.	N/A	N/A
2017/0333200	12/2016	Arnin	N/A	N/A
2017/0367842	12/2016	Predick et al.	N/A	N/A
2017/0367843	12/2016	Eisen et al.	N/A	N/A
2017/0367844	12/2016	Eisen et al.	N/A	N/A
2017/0367845	12/2016	Eisen et al.	N/A	N/A
2018/0000606	12/2017	Hessler et al.	N/A	N/A
2018/0030362	12/2017	Kosler et al.	N/A	N/A
2018/0031810	12/2017	Hsu et al.	N/A	N/A
2018/0036136	12/2017	Duffield et al.	N/A	N/A

2018/0036138	12/2017	Robinson	N/A	N/A
2018/0104066	12/2017	Bae et al.	N/A	N/A
2018/0116819	12/2017	Maguire et al.	N/A	N/A
2018/0116891	12/2017	Beale et al.	N/A	N/A
2018/0193160	12/2017	Hsu et al.	N/A	N/A
2018/0193164	12/2017	Shoshtaev	N/A	N/A
2018/0206999	12/2017	Suddaby	N/A	N/A
2018/0256356	12/2017	Robinson et al.	N/A	N/A
2018/0256359	12/2017	Greenhalgh	N/A	N/A
2018/0256360	12/2017	Cain	N/A	N/A
2018/0256362	12/2017	Slivka et al.	N/A	N/A
2018/0263784	12/2017	Bechtel et al.	N/A	N/A
2018/0271513	12/2017	Perrow et al.	N/A	N/A
2018/0280142	12/2017	Schultz et al.	N/A	N/A
2018/0303473	12/2017	Spann et al.	N/A	N/A
2018/0303621	12/2017	Brotman et al.	N/A	N/A
2018/0303625	12/2017	Alheidt et al.	N/A	N/A
2018/0311048	12/2017	Glerum et al.	N/A	N/A
2018/0318101	12/2017	Engstrom	N/A	N/A
2018/0318102	12/2017	Seifert et al.	N/A	N/A
2018/0325574	12/2017	Bjork et al.	N/A	N/A
2018/0338838	12/2017	Cryder et al.	N/A	N/A
2018/0338841	12/2017	Miller et al.	N/A	N/A
2018/0344307	12/2017	Hynes et al.	N/A	N/A
2018/0360616	12/2017	Luu	N/A	N/A
2019/0000640	12/2018	Weiman	N/A	N/A
2019/0000702	12/2018	Lim et al.	N/A	N/A
2019/0000707	12/2018	Lim et al.	N/A	N/A
2019/0020121	12/2018	Paulotto et al.	N/A	N/A
2019/0021716	12/2018	Waugh et al.	N/A	N/A
2019/0021873	12/2018	Dmuschewsky	N/A	N/A
2019/0046329	12/2018	Padovani et al.	N/A	N/A
2019/0046381	12/2018	Lim et al.	N/A	N/A
2019/0046383	12/2018	Lim et al.	N/A	N/A
2019/0060083	12/2018	Weiman et al.	N/A	N/A
2019/0082949	12/2018	Weiman	N/A	N/A
2019/0083081	12/2018	Ortiz et al.	N/A	N/A
2019/0091033	12/2018	Dewey et al.	N/A	N/A
2019/0105175	12/2018	Zappacosta et al.	N/A	N/A
2019/0125328	12/2018	Blain	N/A	N/A
2019/0133434	12/2018	Lee et al.	N/A	N/A
2019/0133645	12/2018	Gordon et al.	N/A	N/A
2019/0133779	12/2018	McLaughlin et al.	N/A	N/A
2019/0133780	12/2018	Matthews et al.	N/A	N/A
2019/0133784	12/2018	Gunn et al.	N/A	N/A
2019/0133788	12/2018	Weiman et al.	N/A	N/A
2019/0142480	12/2018	Woolley et al.	N/A	N/A
2019/0151115	12/2018	Nichols et al.	N/A	N/A
2019/0183656	12/2018	Stein	N/A	N/A
2019/0201209	12/2018	Branch et al.	N/A	N/A

2019/0201210	12/2018	Besaw et al.	N/A	N/A
2019/0209155	12/2018	Mast et al.	N/A	N/A
2019/0216453	12/2018	Predick et al.	N/A	N/A
2019/0231552	12/2018	Sandul	N/A	N/A
2019/0240039	12/2018	Walker et al.	N/A	N/A
2019/0240043	12/2018	Greenhalgh	N/A	N/A
2019/0247098	12/2018	Brumfield et al.	N/A	N/A
2019/0254650	12/2018	Martinelli et al.	N/A	N/A
2019/0254838	12/2018	Miller et al.	N/A	N/A
2019/0254839	12/2018	Nichols et al.	N/A	N/A
2019/0262009	12/2018	Cheng	N/A	N/A
2019/0262139	12/2018	Wolters	N/A	N/A
2019/0269521	12/2018	Shoshtaev	N/A	N/A
2019/0274670	12/2018	O'Connell et al.	N/A	N/A
2019/0274671	12/2018	Lauf et al.	N/A	N/A
2019/0274836	12/2018	Eisen et al.	N/A	N/A
2019/0282373	12/2018	Alheidt	N/A	N/A
2019/0290446	12/2018	Masson et al.	N/A	N/A
2019/0290447	12/2018	Stein	N/A	N/A
2019/0298416	12/2018	Rezach	N/A	N/A
2019/0298524	12/2018	Lauf et al.	N/A	N/A
2019/0298540	12/2018	Aghayev et al.	N/A	N/A
2019/0321022	12/2018	Karpowicz et al.	N/A	N/A
2019/0321190	12/2018	Wagner et al.	N/A	N/A
2019/0328539	12/2018	Suh et al.	N/A	N/A
2019/0328540	12/2018	Seifert et al.	N/A	N/A
2019/0329388	12/2018	Erickson et al.	N/A	N/A
2019/0336301	12/2018	Engstrom	N/A	N/A
2019/0336304	12/2018	Burkhardt et al.	N/A	N/A
2019/0350573	12/2018	Vogel et al.	N/A	N/A
2019/0358049	12/2018	Faulhaber	N/A	N/A
2019/0358050	12/2018	Fessler	N/A	N/A
2019/0358051	12/2018	Flower et al.	N/A	N/A
2019/0380840	12/2018	Tyber et al.	N/A	N/A
2019/0388232	12/2018	Purcell et al.	N/A	N/A
2020/0008951	12/2019	McClintock et al.	N/A	N/A
2020/0030114	12/2019	Cain	N/A	N/A
2020/0030116	12/2019	Jimenez et al.	N/A	N/A
2020/0038200	12/2019	Foley et al.	N/A	N/A
2020/0054461	12/2019	Marrocco et al.	N/A	N/A
2020/0060844	12/2019	Mathieu et al.	N/A	N/A
2020/0069316	12/2019	DeSoutter et al.	N/A	N/A
2020/0078190	12/2019	Rogers et al.	N/A	N/A
2020/0093526	12/2019	Daly et al.	N/A	N/A
2020/0093607	12/2019	Davenport et al.	N/A	N/A
2020/0093609	12/2019	Shoshtaev	N/A	N/A
2020/0100904	12/2019	Stein et al.	N/A	N/A
2020/0107824	12/2019	Fleischer	N/A	N/A
2020/0129306	12/2019	Miller et al.	N/A	N/A
2020/0129307	12/2019	Hunziker et al.	N/A	N/A

2020/0138591	12/2019	Moskowitz et al.	N/A	N/A
2020/0138593	12/2019	Martynova et al.	N/A	N/A
2020/0146840	12/2019	Black et al.	N/A	N/A
2020/0179120	12/2019	Bielenstein et al.	N/A	N/A
2020/0205993	12/2019	Davenport et al.	N/A	N/A
2020/0214754	12/2019	Bowen et al.	N/A	N/A
2020/0222202	12/2019	Kuyler et al.	N/A	N/A
2020/0229944	12/2019	Suh et al.	N/A	N/A
2020/0246159	12/2019	Suh et al.	N/A	N/A
2020/0246162	12/2019	Schultz et al.	N/A	N/A
2020/0261242	12/2019	Bost et al.	N/A	N/A
2020/0268524	12/2019	Glerum et al.	N/A	N/A
2020/0276028	12/2019	Blain et al.	N/A	N/A
2020/0281741	12/2019	Grotz	N/A	N/A
2020/0289287	12/2019	Emerick et al.	N/A	N/A
2020/0297507	12/2019	Iott et al.	N/A	N/A
2020/0330239	12/2019	Davenport et al.	N/A	N/A
2020/0330245	12/2019	Glerum	N/A	N/A
2020/0345511	12/2019	Daffinson et al.	N/A	N/A
2020/0352731	12/2019	Berry	N/A	N/A
2020/0352738	12/2019	Berry	N/A	N/A
2020/0360153	12/2019	Weiman et al.	N/A	N/A
2020/0375753	12/2019	McLaughlin et al.	N/A	N/A
2020/0375755	12/2019	Cain	N/A	N/A
2020/0383797	12/2019	Predick et al.	N/A	N/A
2020/0383799	12/2019	Cain	N/A	N/A
2020/0390433	12/2019	Yu et al.	N/A	N/A
2020/0390565	12/2019	Jimenez et al.	N/A	N/A
2020/0397593	12/2019	Davenport et al.	N/A	N/A
2020/0405497	12/2019	Olmos et al.	N/A	N/A
2020/0405498	12/2019	Gray et al.	N/A	N/A
2020/0405499	12/2019	Gerbec et al.	N/A	N/A
2020/0405500	12/2019	Cain	N/A	N/A
2021/0007860	12/2020	Glerum et al.	N/A	N/A
2021/0015626	12/2020	Suddaby	N/A	N/A
2021/0030555	12/2020	Weiman et al.	N/A	N/A
2021/0030561	12/2020	Gleason	N/A	N/A
2021/0045891	12/2020	Rogers et al.	N/A	N/A
2021/0045892	12/2020	Rogers et al.	N/A	N/A
2021/0052395	12/2020	Tott et al.	N/A	N/A
2021/0068959	12/2020	McLuen et al.	N/A	N/A
2021/0068974	12/2020	Cowan et al.	N/A	N/A
2021/0068982	12/2020	Cames et al.	N/A	N/A
2021/0077271	12/2020	Sharabani	N/A	N/A
2021/0077272	12/2020	Eisen et al.	N/A	N/A
2021/0085479	12/2020	Weiman et al.	N/A	N/A
2021/0093462	12/2020	Lucasiewicz et al.	N/A	N/A
2021/0106434	12/2020	Alheidt et al.	N/A	N/A
2021/0113349	12/2020	Weiman et al.	N/A	N/A
2021/0121299	12/2020	Hyder	N/A	N/A

2021/0121300	12/2020	Weiman et al.	N/A	N/A
2021/0137697	12/2020	Weiman	N/A	N/A
2021/0137699	12/2020	Jang et al.	N/A	N/A
2021/0137701	12/2020	Miller et al.	N/A	N/A
2021/0154811	12/2020	Spreiter et al.	N/A	N/A
2021/0161678	12/2020	Dewey et al.	N/A	N/A
2021/0177618	12/2020	Branch et al.	N/A	N/A
2021/0186706	12/2020	Spitler et al.	N/A	N/A
2021/0186709	12/2020	Weiman et al.	N/A	N/A
2021/0196470	12/2020	Shoshtaev	N/A	N/A
2021/0205092	12/2020	Glerum et al.	N/A	N/A
2021/0205094	12/2020	Weiman et al.	N/A	N/A
2021/0220145	12/2020	Stein	N/A	N/A
2021/0220147	12/2020	Berry	N/A	N/A
2021/0236298	12/2020	Weiman et al.	N/A	N/A
2021/0251770	12/2020	Purcell et al.	N/A	N/A
2021/0251776	12/2020	Daffinson et al.	N/A	N/A
2021/0259848	12/2020	Kang et al.	N/A	N/A
2021/0259849	12/2020	Robinson et al.	N/A	N/A
2021/0259850	12/2020	Eisen et al.	N/A	N/A
2021/0267767	12/2020	Stein	N/A	N/A
2021/0275317	12/2020	Spetzger	N/A	N/A
2021/0275318	12/2020	Reimels	N/A	N/A
2021/0275319	12/2020	Reimels	N/A	N/A
2021/0275321	12/2020	Seifert et al.	N/A	N/A
2021/0282938	12/2020	Nichols et al.	N/A	N/A
2021/0298915	12/2020	Faulhaber	N/A	N/A
2021/0298916	12/2020	Melkent et al.	N/A	N/A
2021/0307920	12/2020	Walker et al.	N/A	N/A
2021/0315705	12/2020	Altarac et al.	N/A	N/A
2021/0322179	12/2020	Miller et al.	N/A	N/A
2021/0322181	12/2020	Predick	N/A	N/A
2021/0322182	12/2020	Faulhaber	N/A	N/A
2021/0330472	12/2020	Shoshtaev	N/A	N/A
2021/0346174	12/2020	Flint et al.	N/A	N/A
2021/0353277	12/2020	Gregersen et al.	N/A	N/A
2021/0401586	12/2020	Zakelj	N/A	A61F 2/4455
2022/0015924	12/2021	Freedman et al.	N/A	N/A
2022/0047312	12/2021	Seykora et al.	N/A	N/A
2022/0087819	12/2021	Robinson et al.	N/A	N/A
2022/0133336	12/2021	Tsai et al.	N/A	N/A
2022/0133497	12/2021	Dewey et al.	N/A	N/A
2022/0133498	12/2021	Josse et al.	N/A	N/A
2022/0133499	12/2021	Josse et al.	N/A	N/A
2022/0218325	12/2021	Josse	N/A	N/A
2022/0265256	12/2021	Villamil et al.	N/A	N/A
2022/0313450	12/2021	Donohoe et al.	N/A	N/A
2022/0387184	12/2021	Josse et al.	N/A	N/A
2022/0409397	12/2021	Hayes et al.	N/A	N/A
2023/0015512	12/2022	Eisen et al.	N/A	N/A

2023/0027836	12/2022	Predick et al.	N/A	N/A
2023/0137358	12/2022	Hayes et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
107 137 166	12/2016	CN	N/A
44 16 605	12/1994	DE	N/A
0 767 636	12/1996	EP	N/A
0 880 950	12/1997	EP	N/A
0 857 042	12/2000	EP	N/A
1 442 732	12/2003	EP	N/A
1 124 512	12/2003	EP	N/A
1 107 711	12/2003	EP	N/A
1 506 753	12/2004	EP	N/A
1 459 711	12/2006	EP	N/A
2954860	12/2014	EP	N/A
3031424	12/2015	EP	N/A
3 069 694	12/2015	EP	N/A
3213720	12/2016	EP	N/A
2781998	12/1999	FR	N/A
3082115	12/2018	FR	N/A
2 377 387	12/2002	GB	N/A
102117224	12/2019	KR	N/A
102192022	12/2019	KR	N/A
92/14423	12/1991	WO	N/A
97/ 00054	12/1996	WO	N/A
99/26562	12/1998	WO	N/A
99/66867	12/1998	WO	N/A
00/12033	12/1999	WO	N/A
00/25706	12/1999	WO	N/A
00/49977	12/1999	WO	N/A
02/19952	12/2001	WO	N/A
03/105673	12/2002	WO	N/A
2006116850	12/2005	WO	N/A
2012139022	12/2011	WO	N/A
2014/133755	12/2013	WO	N/A
2015063721	12/2014	WO	N/A
2015198335	12/2014	WO	N/A
2016057940	12/2015	WO	N/A
2016/205607	12/2015	WO	N/A
2017/168208	12/2016	WO	N/A
2018049227	12/2017	WO	N/A
2021055323	12/2020	WO	N/A

OTHER PUBLICATIONS

Chinese Office Action in Application No. 201980010758.4 dated Sep. 16, 2023. cited by applicant
International Search Report and Written Opinion in Application No. PCT/US2022/030094 dated Sep. 16, 2022. cited by applicant
International Search Report, and Written Opinion for Application. No. PCT/US2019/019067, dated

Jun. 3, 2019. cited by applicant
International Search Report and Written Opinion for Application No. PCT/US2019/019060, dated Jun. 5, 2019. cited by applicant
International Search Report and Written Opinion, PCT/IB2020/000932, dated Jul. 29, 2021. cited by applicant
International Search Report and Written Opinion, PCT/IB2020/000942, dated Aug. 10, 2021. cited by applicant
European Search Report, EP19756905, dated Oct. 18, 2021. cited by applicant
International Search Report and Written Opinion in Application No. PCT/IB2023/058417 dated Dec. 7, 2023. cited by applicant
International Search Report and Written Opinion in Application No. PCT/IB2023/057720 dated Nov. 8, 2023. cited by applicant
International Search Report and Written Opinion in Application No. PCT/US2022/027200 dated Aug. 19, 2022. cited by applicant
International Search Report and Written Opinion in Application No. PCT/US2022/016831 dated Sep. 29, 2022. cited by applicant
Chinese Office Action in Application No. 201980010758.4 dated Jun. 16, 2023. cited by applicant
International Search Report and Written Opinion in Application No. PCT/US2022/016809 dated Jul. 27, 2022. cited by applicant
International Search Report and Written Opinion in Application No. PCT/US2022/027695 dated Jul. 27, 2022. cited by applicant
European Search Report in Application No. 22828961.7 dated Mar. 14, 2025. cited by applicant
International Search Report and Written Opinion in Application No. PCT/IB2024/054985 dated Sep. 10, 2024. cited by applicant

Primary Examiner: Weiss; Jessica

Attorney, Agent or Firm: FOX ROTHSCCHILD LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation in part of: U.S. patent application Ser. No. 17/665,449, titled EXPANDABLE IMPLANT AND CORRESPONDING INSERTER, filed Feb. 4, 2022; U.S. patent application Ser. No. 17/515,709, titled EXPANDABLE IMPLANT AND CORRESPONDING INSERTER, filed Nov. 1, 2021; and U.S. patent application Ser. No. 17/356,950, titled EXPANDABLE INTERBODY IMPLANT, filed Jun. 24, 2021 the entire disclosure of each application is incorporated herein by reference. This application also incorporates by reference U.S. application Ser. No. 17/307,578, titled EXTERNALLY DRIVEN EXPANDABLE INTERBODY AND RELATED METHODS, filed May 5, 2021; U.S. Pat. No. 11,096,796, titled INTERBODY SPINAL IMPLANT HAVING A ROUGHENED SURFACE TOPOGRAPHY ON ONE OR MORE INTERNAL SURFACES, and filed on Mar. 4, 2013; and U.S. Pat. No. 10,821,000, titled TITANIUM IMPLANT SURFACES FREE FROM ALPHA CASE AND WITH ENHANCED OSTEOINDUCTION, and filed Jun. 29, 2017.

FIELD

(1) In one aspect, the present technology is generally related to an externally driven expandable interbody implant for use in a medical procedure related to the spine. In some embodiments, disclosed implants may be used in an anterior cervical discectomy and fusion (ACDF) procedure

although other uses in other areas of the spine or between two bones are also contemplated.

BACKGROUND

(2) Mechanically operated interbody implants may be used to align and/or realign a patient's spine during a medical procedure and/or for purposes of fusion, degenerative tissue and/or trauma/repair procedures. Conventional implants designed for the Thoracic and Lumbar region of the spine often include top and bottom endplates and a mechanical means to separate the top and bottom endplates. The mechanical mechanisms to separate the top and bottom endplates are often cumbersome and require a large footprint that is often unsuitable, for example, for ACDF type surgeries of the cervical portion of the spine. Many currently available ACDF type implants may be limited in an ability to optimize the adjustment of lordosis or sagittal alignment of the vertebral bodies because they may rely on a fixed lordotic angle between the superior/cephalad and inferior/caudad faces of the device.

SUMMARY

(3) The techniques of this disclosure generally relate to an expandable interbody implant including a superior endplate and an inferior endplate hingedly coupled and which may further include a locking element to secure the inferior endplate and superior endplate in a particular configuration, for example. The superior and inferior endplates may be moved in a multitude of expanded and/or lordosed or kyphosed or otherwise angled configurations via an external inserter for example. In various embodiments, a locking screw may be a breakoff type screw. In various embodiments at least one breakoff tang on the implant may be used to for gripping of the implant to insert it into a disc space and afterwards the breakoff tang may be broken off and removed. Additionally, in various embodiments the locking screw may be used to grip the implant and insert it into a disc space. Additionally, in various embodiments female recesses, rather than tangs (or male bosses/protrusions), may be used for gripping of the implant and inserting the implant into a disc space.

(4) In one aspect, the present disclosure provides for an expandable implant movable between a contracted position (closed position) and an expanded position, for example. The expandable implant may include an expandable body extending from a proximal end to a distal end in a proximal-to-distal direction (may also be referred to as an anterior-to-posterior direction depending on surgical technique), extending from a first lateral side to a second lateral side in a widthwise direction, and extend from a superior endplate to an inferior endplate in a heightwise direction (may also be referred to as a cephalad-to-caudal and/or vertical direction depending on surgical technique), for example. In various embodiments, the expandable body may be defined by a superior endplate and an inferior endplate that are hingedly connected, for example. In various embodiments, the superior endplate includes a first core having a distal engagement surface (may also be referred to as a posterior engagement surface depending on surgical technique) and the inferior endplate includes a second core having a proximal engagement surface (may also be referred to as an anterior engagement surface) and a threaded screw aperture, for example. In various embodiments, disclosed implants may include a threaded breakoff screw having a fracture surface that is disposed in the threaded screw aperture and movable between a locked position and an unlocked position, for example. In various embodiments, when in the locked position, the breakoff screw urges the distal engagement surface of the first core into direct contact with the proximal engagement surface of the second core, for example.

(5) In another aspect, the disclosure provides for a system including a medical implant and a surgical tool, for example. The system may include an expandable implant movable between a contracted position and an expanded position, for example. In various embodiments, the expanded position may also refer to a distracted and angled orientation of the superior endplate and inferior endplate. The expandable implant may include an expandable body extending from a proximal end to a distal end in a proximal-to-distal direction and extending from a first lateral side to a second lateral side in a widthwise direction, for example. In various embodiments, the expandable body

may be defined by a superior endplate and an inferior endplate that are hingedly connected, for example. In various embodiments, the superior endplate includes a first core having a distal engagement surface and the inferior endplate includes a second core having a proximal engagement surface and a threaded screw aperture, for example. In various embodiments, disclosed implants may include a locking screw that is disposed in the threaded screw aperture and movable between a locked position and an unlocked position, for example. In various embodiments, when in the locked position, the locking screw urges the distal engagement surface of the first core into direct contact with the proximal engagement surface of the second core, for example. The system may also include a surgical tool for expanding the implant and tightening the locking screw while the implant is expanded at a desired height, position, and/or angle.

(6) The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view of an expandable implant.
- (2) FIG. 2 is an alternate perspective view of an expandable implant.
- (3) FIG. 3 is a top down view of an expandable implant.
- (4) FIG. 4 is a side view of an expandable implant.
- (5) FIG. 5 is a rear perspective view of an expandable implant.
- (6) FIG. 6 is a perspective view of the interior of a superior endplate of an expandable implant.
- (7) FIG. 7 is a perspective view of the interior of an inferior endplate of an expandable implant.
- (8) FIG. 8 is a perspective exploded parts view of an expandable implant.
- (9) FIG. 9 is an exploded parts view of an expandable implant from a side view perspective.
- (10) FIG. 10 is a perspective cross section view of an expandable implant.
- (11) FIG. 11 is a cross section view of an expandable implant.
- (12) FIG. 12 is a side view of a superior endplate for use with at least some expandable implant embodiments.
- (13) FIG. 13A is a perspective view of a first expandable implant.
- (14) FIG. 13B is a perspective view of a second expandable implant.
- (15) FIG. 14A is a perspective view of a third expandable implant.
- (16) FIG. 14B is a perspective view of a fourth expandable implant.
- (17) FIG. 15 is a perspective view of a fifth expandable implant.
- (18) FIG. 16 is a side view of an expandable implant in the expanded configuration.
- (19) FIG. 17 is a side view of an expandable implant showing a bone screw trajectory.
- (20) FIG. 18 is a front view of an expandable implant.
- (21) FIG. 19 is a front view of an enlarged area of FIG. 18.
- (22) FIG. 20 is a perspective view of an inserter for use with disclosed expandable implants.
- (23) FIG. 21 is a perspective view of an inserter for use with disclosed expandable implants shown in skeleton outlining for ease of understanding.
- (24) FIG. 22 is a perspective view of an inserter for use with disclosed expandable implants shown in skeleton outlining for ease of understanding.
- (25) FIG. 23A is a rear view of an inserter in a non-expanded position.
- (26) FIG. 23B is a rear view of an inserter in an expanded position.
- (27) FIG. 24 is an enlarged view of a distal end of an inserter in an expanded position coupled to an example expandable implant in a corresponding expanded position.
- (28) FIG. 25 is a perspective view of an expandable implant in an expanded configuration after a

breakoff portion of a locking screw has been broken off.

(29) FIG. **26** is a perspective view of a surgical instrument for use with disclosed expandable implants.

(30) FIG. **27** is a perspective view of a surgical instrument for use with disclosed expandable implants.

(31) FIG. **28** is a perspective view of an expandable implant after the mounting tangs have been broken off.

(32) FIG. **29** is a reference drawing showing the human spine of which various disclosed implant embodiments may be installed in.

(33) FIG. **30** is a reference drawing showing various planes and reference directions of which the various disclosed implant embodiments may move in or act in with respect to a patient.

(34) FIG. **31** is a perspective view of a second implant embodiment.

(35) FIG. **32** is a first perspective exploded parts view of the second implant embodiment.

(36) FIG. **33** is a second perspective exploded parts view of the second implant embodiment.

(37) FIG. **34** is a side exploded parts view of the second implant embodiment.

(38) FIG. **35** is a first side view of a breakoff screw having a recessed fracture surface.

(39) FIG. **36** is a second side view of a breakoff screw having a recessed fracture surface.

(40) FIG. **37** is a cross section view of a breakoff screw having a recessed fracture surface.

(41) FIG. **38** is a perspective view of a swaging fixture.

(42) FIG. **39** is a cross section view of a swage mandrel and a distal end of a breakoff screw before the commencement of a swage process.

(43) FIG. **40** is a cross section view showing a result of a swage process.

(44) FIG. **41** is an enlarged view of region S-W of FIG. **40**.

(45) FIG. **42A** is a front perspective view of a third implant embodiment.

(46) FIG. **42B** is an alternate front perspective view of a third implant embodiment.

(47) FIG. **43A** is a front perspective view of an implant having angled engagement features.

(48) FIG. **43B** is a top down view of an implant having angled engagement features.

(49) FIG. **44** is a perspective view of a surgical tool for use with disclosed implant embodiments.

(50) FIG. **45** is a first side view of a surgical tool for use with disclosed implant embodiments.

(51) FIG. **46A** is a second side view of a surgical tool for use with disclosed implant embodiments.

(52) FIG. **46B** is a third side view of a surgical tool in an operative position for use with disclosed implant embodiments.

(53) FIG. **46C** is a perspective view of a surgical tool in the operative position.

(54) FIG. **47A** is first exploded parts view of a surgical tool for use with disclosed implant embodiments.

(55) FIG. **47B** is second exploded parts view of a surgical tool for use with disclosed implant embodiments.

(56) FIG. **48** is a perspective view showing a surgical tool immediately prior to coupling with a surgical implant.

(57) FIG. **49** is a first cross section drawing of the surgical tool and implant in a coupled configuration.

(58) FIG. **50** is a second cross section drawing of the surgical tool and implant in a coupled configuration.

(59) FIG. **51** is a third cross section drawing of the surgical tool and implant in a coupled configuration.

DETAILED DESCRIPTION

(60) Embodiments of the present disclosure relate generally, for example, to spinal stabilization systems, and more particularly, to surgical instruments for use with spinal stabilization systems. Embodiments of the devices and methods are described below with reference to the Figures.

(61) The following discussion omits or only briefly describes certain components, features and

functionality related to medical implants, installation tools, and associated surgical techniques, which are apparent to those of ordinary skill in the art. It is noted that various embodiments are described in detail with reference to the drawings, in which like reference numerals represent like parts and assemblies throughout the several views, where possible. Reference to various embodiments does not limit the scope of the claims appended hereto because the embodiments are examples of the inventive concepts described herein. Additionally, any example(s) set forth in this specification are intended to be non-limiting and set forth some of the many possible embodiments applicable to the appended claims. Further, particular features described herein can be used in combination with other described features in each of the various possible combinations and permutations unless the context or other statements clearly indicate otherwise.

(62) Terms such as “same,” “equal,” “planar,” “coplanar,” “parallel,” “perpendicular,” etc. as used herein are intended to encompass a meaning of exactly the same while also including variations that may occur, for example, due to manufacturing processes. The term “substantially” may be used herein to emphasize this meaning, particularly when the described embodiment has the same or nearly the same functionality or characteristic, unless the context or other statements clearly indicate otherwise.

(63) Referring to FIGS. **1-43** generally, various embodiments and views of an expandable implant **100** are disclosed. The components of expandable implant **100** can be fabricated from biologically acceptable materials suitable for medical applications, including metals, synthetic polymers, ceramics and bone material and/or their composites. For example, the components, individually or collectively, can be fabricated from materials such as stainless steel alloys, commercially pure titanium, titanium alloys, Grade 5 titanium, super-elastic titanium alloys, cobalt-chrome alloys, superelastic metallic alloys (e.g., Nitinol, super elasto-plastic metals, such as GUM METAL™), ceramics and composites thereof such as calcium phosphate (e.g., SKELITE™), thermoplastics such as polyaryletherketone (PAEK) including polyetheretherketone (PEEK), polyetherketoneketone (PEKK) and polyetherketone (PEK), carbon-PEEK composites, PEEK-BaSO₄ polymeric rubbers, polyethylene terephthalate (PET), fabric, silicone, polyurethane, silicone-polyurethane copolymers, polymeric rubbers, polyolefin rubbers, hydrogels, semi-rigid and rigid materials, elastomers, rubbers, thermoplastic elastomers, thermoset elastomers, elastomeric composites, rigid polymers including polyphenylene, polyamide, polyimide, polyetherimide, polyethylene, epoxy, bone material including autograft, allograft, xenograft or transgenic cortical and/or corticocancellous bone, and tissue growth or differentiation factors, partially resorbable materials, such as, for example, composites of metals and calcium-based ceramics, composites of PEEK and calcium based ceramics, composites of PEEK with resorbable polymers, totally resorbable materials, such as, for example, calcium based ceramics such as calcium phosphate, tri-calcium phosphate (TCP), hydroxyapatite (HA)-TCP, calcium sulfate, or other resorbable polymers such as polyaetide, polyglycolide, polytyrosine carbonate, polycaroplaetohe, polylactic acid or polylactide and their combinations.

(64) In various embodiments, components may be coated with a ceramic, titanium, and/or other biocompatible material to provide surface texturing at (a) the macro scale, (b) the micro scale, and/or (c) the nano scale, for example. Similarly, components may undergo a subtractive manufacturing process providing for surface texturing configured to facilitate osseointegration and cellular attachment and osteoblast maturation. Example surface texturing of additive and subtractive manufacturing processes may comprise (a) macro-scale structural features having a maximum peak-to-valley height of about 40 microns to about 500 microns, (b) micro-scale structural features having a maximum peak-to-valley height of about 2 microns to about 40 microns, and/or (c) nano-scale structural features having a maximum peak-to-valley height of about 0.05 microns to about 5 microns. In various embodiments, the three types of structural features may be overlapping with one another, for example. Additionally, such surface texturing may be applied to any surface, e.g., both external exposed facing surfaces of components and internal non

exposed surfaces of components. Further discussion regarding relevant surface texturing and coatings is described in, for example, U.S. Pat. No. 11,096,796, titled Interbody spinal implant having a roughened surface topography on one or more internal surfaces, and filed on Mar. 4, 2013—the entire disclosure of which is incorporated herein by reference in its entirety. Accordingly, it shall be understood that any of the described coating and texturing processes of U.S. Pat. No. 11,096,796, may be applied to any component of the various embodiments disclosed herein, e.g., the exposed surfaces and internal surfaces of endplates. Another example technique for manufacturing an orthopedic implant having surfaces with osteoinducting roughness features including micro-scale structures and nano-scale structures is disclosed in U.S. Pat. No. 10,821,000, the entire contents of which are incorporated herein by reference. Additionally, an example of a commercially available product may be the Adaptix™ Interbody System sold by Medtronic Spine and comprising a titanium cage made with Titan nanoLOCK™.

(65) Referring generally to FIGS. 1-5 various views of an expandable implant **100** in a collapsed position are illustrated. FIGS. 1-2 are various perspective views of an expandable implant **100**. FIG. 3 is a top down view of an expandable implant **100**. In the example embodiment, expandable implant **100** may include a proximal end **100P**, a distal end **100D**, and first and second lateral sides **100L**. Additionally, a pair of bone screw apertures **11**, **21** may be positioned on the proximal end **100P**, for example. In various embodiments, bone screw apertures **11**, **21** may comprise a corresponding bone screw retention mechanism **11a**, **21a** (may also be referred to as an anti-backout locking mechanism). In the example embodiment, the bone screw retention mechanisms **11a**, **21a**, comprise a flexible tang member having a hook portion at an end thereof that allows the flexible tang member to flex outward in a lateral direction away from the corresponding bone screw aperture **11**, **21** during initial installation of the bone screw and to flex back inward towards the corresponding bone screw aperture **11**, **21** to prevent a corresponding bone screw from backing out. For example, as the bone is installed in bone screw aperture **11**, **21**, the bone screw retention mechanism **11a**, **21a**, may flex outward as the underside of the head portion contacts inclined surface **11c** (see FIG. 19).

(66) In various embodiments, and as illustrated in FIGS. 1-2, mounting tangs **19**, **29** may extend in a proximal direction (may also be referred to as an anterior direction depending on surgical technique and orientation), for example. In various embodiments, implant **100** may be referred to as an externally driven expandable implant because an end user or surgeon may use a surgical tool to open and close implant **100**, e.g. expand implant **100**. For example, an external tool may adjust the lordotic angle of implant **100** as will be explained in detail with respect to FIGS. 20-25. Once implant **100** is expanded to an appropriate lordotic angle (also referred to as angle of inclination), an end user may fix the relative angle of the superior endplate **10** relative to the inferior endplate **20** by tightening locking screw **50**, for example. In some embodiments, superior endplate **10** may be referred to as a “cephalad” endplate and inferior endplate **20** may be referred to as a “caudal” endplate.

(67) Locking screw **50** may also be used in other embodiments, such as fixation of posterior rods, fixation of pedicle screws, and other set screw constructs. Additionally, locking screw **50** may be referred to as a “breakoff screw” in some embodiments.

(68) At least one advantage of relying on an external tool to adjust a lordotic angle of implant **100** may be the reduction of internal components within implant **100** relative to other forms of implants relying on various moving mechanisms and/or expansion mechanisms, for example. Accordingly, in various embodiments, implant **100** may have a relatively large void space in the interior thereof, which may facilitate a fusion process during an ACDF procedure. For example, implant **100** may have a relatively large internal volume **101** that is open through the superior endplate **10** and inferior endplate **20** which may be packed with bone graft material, for example.

(69) As illustrated in FIG. 3, implant **100** may extend in a proximal-to-distal direction (may also be referred to as a longitudinal direction and/or an anterior-to-posterior direction depending on

surgical technique and final orientation) from the proximal end **100P** (may be referred to as anterior end depending on surgical technique and final orientation) to the distal end **100D** (may be referred to as posterior end depending on surgical technique and final orientation) through axis P-D through the center of the implant **100**, for example. Implant **100** may extend in a widthwise direction (also referred to as lateral direction) from the first lateral side **100L** to the second lateral side **100L** through axis W-W through the center of the implant **100** and the center of locking screw **50**, for example. The axis P-D may be perpendicular and/or substantially perpendicular to the axis W-W. For example, the proximal-to-distal direction may be perpendicular to the widthwise direction. Additionally, a width of the implant may taper from a proximal end **100P** where it is widest towards a distal end **100D** where it is narrowest. In various embodiments, implant **100** may extend from a superior endplate **10** to an inferior endplate **20** in a heightwise direction (may also be referred to as a cephalad-to-caudal and/or vertical direction depending on surgical technique and final orientation).

(70) FIG. 4 is a side view of an expandable implant **100**. In the example illustration, it is shown that a superior endplate **10** is connected to an inferior endplate **20** such that the superior endplate may pivot about a hinge member **40**. In the example embodiment, hinge member **40** comprises an arcuate rail portion of inferior endplate **20** that extends in the widthwise direction, for example. In other embodiments, the hinge member **40** may be reversed relative to the superior and inferior endplates than as illustrated in FIG. 4. In the example embodiment, hinge member **40** may be nested into a corresponding arcuate cavity of the superior endplate **10** such that superior endplate **10** may expand and/or otherwise rotate about hinge member **40**. Additionally, in various embodiments the superior endplate **10** and/or inferior endplate **20** may include various engagement elements **14** for engaging with an adjacent boney structure such as a vertebrae, for example. In the example embodiment, the engagement elements comprise a series of alternating rails and valleys therebetween that extend in the widthwise direction. In some embodiments, the valleys may be angled about 20-40 degrees and in at least one embodiment the valleys may be angled at about 30 degrees relative to the corresponding rail (see FIGS. 42A and 42B). At least one advantage of this orientation may be a relatively greater resistance and/or suppression of expulsion of the implant **100** in both the lateral direction and in the proximal-to-distal direction. However, claws, hooks, dimples, spikes, etc. are also contemplated as example engagement elements **14**. In some embodiments, an acid etching process may be utilized to form a roughened or textured surface to facilitate securing the implant between boney portions and/or suppressing expulsion of implant **100**.

(71) FIG. 5 is a rear perspective view of an expandable implant **100**. In the example illustration it is shown that the distal end **100D** is narrower than the proximal end **100P**. FIG. 6 is a perspective view of the interior of a superior endplate **10**. In the example illustration, it is shown that the distal end of superior endplate **10** includes an arcuate channel **12** of which the hinge member **40** may be disposed inside of. The proximal end of superior endplate **10** may include a bone screw aperture cutout **21b** to allow a relief area for a corresponding bone screw to be insert through bone screw aperture **21** of inferior endplate **20**, for example. Superior endplate **10** may also include a core **15** comprising an aperture **15a** that extends from a proximal surface **15p** thereof to a distal surface **15d** thereof, for example. In various embodiments, aperture **15a** may be referred to as a “slot” or “screw slot”. In some embodiments, core **15** may be referred to as a support frame and take a generally rectangular shape. In various embodiments, the distal surface **15d** may be curved and generally face the distal end **100D** of implant **100**. FIG. 7 is a perspective view of the interior of an inferior endplate **20**. In the example illustration, it is shown that the distal end of inferior endplate **20** includes a hinge member **40** in the form of an arcuate rail that may be disposed inside of the arcuate channel **12** of the superior endplate **10**, for example. The proximal end of inferior endplate **20** may include a bone screw aperture cutout **11b** to allow a relief area for a corresponding bone screw to be insert through bone screw aperture **11** of superior endplate **10**, for example. Inferior

endplate **20** may also include a core **25** comprising a threaded aperture **25a** that extends from a proximal surface **25p** thereof to a distal surface **25d** thereof, for example. In some embodiments, core **25** may be referred to as a support frame and take a generally rectangular shape. In various embodiments, the superior endplate **10** and inferior endplate **20** may each be formed by a unitary single piece, respectively.

(72) FIG. **8** is a perspective exploded parts view and FIG. **9** is an exploded parts view from a side view perspective of an expandable implant **100**. In the example embodiment, a locking screw **50** is illustrated. Locking screw **50** may include an external thread pattern **51** on an outside circumferential surface thereof, for example. The external thread pattern **51** of locking screw **50** may have a size and shape generally corresponding to the threaded aperture **25a** of core **25** of inferior endplate **20**, for example. In various embodiments, an engagement surface **54** may be disposed adjacent and proximal of external thread pattern **51**. In the example embodiment, engagement surface **54** is shaped like a washer and is directly connected to locking screw **50**. However, in other embodiments, engagement surface **54** may be a washer or separated element, for example. In some embodiments, engagement surface **54** may be conically shaped and/or spherically shaped. Engagement surface **54** may include a relatively planar and/or flat distal surface and/or proximal surface (anterior/ventral surface). In various embodiments, a proximal end of set screw **50** may include an aperture having an internal threaded surface **52**. For example, a cylindrical shaped proximal end may include an aperture having a thread pattern disposed on an internal circumferential surface of the cylindrical shaped proximal end. In the example embodiment, a first drive feature **53a** and a second drive feature **53b** may be disposed adjacent to and distally with respect to a proximal most end of set screw **50**. Additionally, first and second drive features **53a**, **53b** may be disposed proximally with respect to engagement surface **54**. In the example embodiment, drive features **53a**, **53b** take a hexalobular shape, although various other shapes such as hexagonal, polygonal, Torx, etc. are also contemplated. In some embodiments, a surgical drive tool having a corresponding socket may be coupled to drive features **53a**, **53b** to cause rotation of locking screw **50**. Similarly, in some alternative embodiments, a drive tool with a protruding threaded member having a thread pattern with a corresponding size and shape to internal threaded surface **52** may also cause rotation of set screw **50**.

(73) As seen best in FIG. **9**, set screw **50** may also include a breakoff location **55**, for example. In the example embodiment, breakoff location **55** is disposed directly between drive features **53a**, **53b** and is designed to shear off when a sufficient rotational force (torque) is applied to a proximal end of set screw **50** while a distal end of set screw **50** is stationary, e.g., when set screw **50** is secured in a locked position and a continued rotational force (torque) is applied to the proximal end of set screw **50** the drive feature **53a** and cylindrical end having the internal threaded surface **52** may breakoff. In various embodiments, the internal threaded surface **52** may also be utilized to ensure that the broken off portion is removed from the patient and remains connected to a surgical tool/breakoff tool. As also seen best in FIG. **9**, the inferior endplate **20** may include a first relief **40a** and a second relief **40b** on opposite sides of hinge member **40**. The first relief **40a** may have a size and shape corresponding to a size and shape of a first portion **12a** of superior endplate **10** and the second relief **40b** may have a size and shape corresponding to a size and shape of a second portion **12b** of superior endplate **10**, for example. In various embodiments, portions **12a**, **12b** may comprise a hook shape, outdent, and/or protrusion, for example. In the example embodiment, portions **12a**, **12b** may be disposed on opposite sides of channel **12** and cup hinge member **40** such that the superior endplate **10** and inferior endplate **20** may rotate relative to one another without becoming uncoupled.

(74) FIG. **10** is a perspective cross section view and FIG. **11** is a cross section view of expandable implant **100**. In the example embodiment, the superior endplate **10** and inferior endplate **20** are coupled together by hinge member **40**, and core **25** may be positioned behind of core **15**, e.g., core **25** may be positioned distally with respect to core **15**. Additionally, the outside external thread

pattern 51 of locking screw 50 may engage with the threaded aperture 25a of the core 25 and extend through aperture 15a of core 15. In this way, when locking screw 50 is rotated, the distal surface 15d of core 15 may engage with the proximal surface 25p of core 25. For example, by tightening locking screw 50 the engagement surface 54 of locking screw 50 pushes against the proximal surface 15p of core 15 thereby bringing the superior endplate 10 and inferior endplate 20 into frictional engagement.

(75) FIG. 12 is a side view of a superior endplate 10 for use with at least some expandable implant 100 embodiments. In the example embodiment, superior endplate 10 may include an arcuate channel 12 of which the hinge member 40 may be disposed inside of. In various embodiments, arcuate channel 12 may be defined by a first circle having a center point at P.sub.1 and/or a segment of the circle having the center point at P.sub.1. The center point P.sub.1 may define an axis of rotation that superior endplate 10 may rotate and/or pivot with respect to inferior endplate 20. For example, superior endplate 10 may be hingedly coupled to hinge member 40 as explained above and rotatable about an axis of rotation defined by center point P.sub.1, for example. Additionally, in various embodiments a distal surface 15d of core 15 may be a curved surface defined (in part or in total) by a second circle having a center point at P.sub.1 and a radius R.sub.1. The proximal surface 15p of core 15 may also be a curved surface defined (in part or in total) by a segment of a circle having a radius R.sub.2 and a center point P.sub.2. In the example embodiment, P.sub.2 is located a distance D.sub.1 above point P.sub.1 and radius R.sub.2 is greater than radius R.sub.1. Additionally, the proximal surface 15p of core 15 is offset a distance D.sub.2 from the proximal most face of the superior endplate 10. In the example embodiment, center point P.sub.2 is vertically above center point P.sub.1 however, in other embodiments, center point P.sub.2 may be offset by a greater amount or even a lesser amount than illustrated. In some examples, P.sub.2 may not be aligned vertically above P.sub.1. In various embodiments, R.sub.1 may be about 7-9 mm+/-about 1 mm and R.sub.2 may be about 8-10 mm+/-about 1 mm although these numbers may be modified in some embodiments having a larger or smaller footprint. In various embodiments D.sub.1 is about 0.25 mm to about 1.0 mm and D.sub.2 is about 0.25 mm to about 1.25 mm. In at least one embodiment, D.sub.1 is about 0.75 mm and D.sub.2 is about 0.8 mm and R.sub.2 is about 9.2 mm.

(76) The above explained geometrical relationship between the offset center points P.sub.1 and P.sub.2 and R.sub.1 and R.sub.2 may have several advantages in terms of operability and functionality. At least one advantage is that the superior endplate 10 may have a natural tendency to apply a force against the engagement surface 54 of locking screw 50 such that locking screw 50 may function similar to a wedge preventing implant 100 from fully collapsing. Another advantage is that a biasing force may be applied that naturally urges the superior endplate 10 and inferior endplate 20 into an expanded position which may assist with expanding the implant 100 when positioned between a superior vertebrae and an inferior vertebrae, for example. For example still, an end user such as a surgeon may expand implant 100 and the offset arrangement explained above may facilitate the function of keeping implant 100 lordosed at the chosen angle.

(77) FIG. 13A is a perspective view of a first expandable implant, FIG. 13B is a perspective view of a second expandable implant, FIG. 14A is a perspective view of a third expandable implant, FIG. 14B is a perspective view of a fourth expandable implant, and FIG. 15 is a perspective view of a fifth expandable implant. In the series of illustrations it is shown that various embodiments in accordance with the principles of this disclosure may be variously sized depending on the particular location in a human body and the particular patient specific human anatomy. For example, FIG. 13A illustrates a first expandable implant 100 having a first height H.sub.1 or thickness between the superior endplate 10 and inferior endplate 20, FIG. 13B illustrates a second expandable implant 100 having a second height H.sub.2 or thickness, FIG. 14A illustrates a third expandable implant 100 having a third height H.sub.3 or thickness, FIG. 14B illustrates a fourth expandable implant 100 having a fourth height H.sub.4 or thickness, and FIG. 15 illustrates a fifth expandable implant

100 having a fifth height $H_{sub.5}$ or thickness. In at least some embodiments, $H_{sub.1}$ may be about 5 mm, $H_{sub.2}$ may be about 6 mm, $H_{sub.3}$ may be about 7 mm, $H_{sub.4}$ may be about 8 mm, $H_{sub.5}$ may be about 9 mm, for example. In various embodiments, an angle of inclination between the superior endplate **10** and inferior endplate **20** may be about 4 degrees to about 15 degrees in an expanded configuration, e.g., an angled and/or inclined configuration.

(78) FIG. **16** is a side view of an expandable implant **100** in the expanded configuration. In an expanded position, a distance $D_{sub.3}$ between the superior endplate **10** and inferior endplate **20** at the proximal end **100P** may be relatively greater than in the closed configuration, for example. Additionally, an angle of inclination α may be relatively greater in an expanded position than in the closed configuration, for example. In this embodiment, implant **100** may have a height $H_{sub.1}$ corresponding to FIG. **13A** and be about 5 mm in a closed configuration. In the illustrated expanded configuration of FIG. **16**, $D_{sub.3}$ may be about 8 mm to 9 mm and α may be about 10 degrees to about 20 degrees. In at least one embodiment, $D_{sub.3}$ may be 8 mm in a fully expanded position and α may be about 15 degrees, for example.

(79) FIG. **17** is a side view of an expandable implant **100** showing a bone screw trajectory **99**. In the example embodiment, it is shown that a centered bone screw trajectory **99** of bone screw **97** is at an angle β with respect to a plane **98** that crosses through a center of the implant from a first lateral side to a second lateral side, for example. Additionally, the bone screw trajectory **99** may be varied $\pm$ by a degree γ , for example. In various embodiments, β may be about 30 degrees to about 50 degrees and γ may be about 2 degrees to about 10 degrees. In the example embodiment, β may be about 40 degrees and γ may be about 5 degrees.

(80) FIG. **18** is a front view of an expandable implant showing an area $A_{sub.1}$ and FIG. **19** is a front view of an enlarged area $A_{sub.1}$ of FIG. **18**. In the example embodiment, bone screw **97** is in a position extending through bone screw aperture **11** where it cannot backout due to bone screw retention mechanism **11a**. The bone screw retention mechanism **11a** includes an inclined surface **11c** such that when bone screw **97** is being installed, an underside of the head portion of bone screw **97** directly contacts the inclined surface **11c** thereby pushing the bone screw retention mechanism **11a** laterally outward and away from bone screw aperture **11**, for example. Thereafter, when bone screw **97** is installed and the head portion of bone screw **97** is beneath inclined surface **11c** the bone screw retention mechanism may flex back towards bone screw aperture **11** such that it will prevent bone screw **97** from backing out, e.g., a blocking surface of bone screw retention mechanism **11a** may contact an upper surface of the head portion of bone screw **97**. In the example embodiment, bone screw retention mechanism **11a** comprises flexible arm (or spring tab) having an inclined surface **11c** (or ramp) that is disposed on a lateral end of implant **100** adjacent bone screw aperture **11**.

(81) Referring generally to FIGS. **20-24** an inserter **200** for use with disclosed expandable implants **100** is illustrated. Inserter **200** may extend in a longitudinal direction from a proximal end **200p** to a distal end **200d**, for example. Inserter **200** may include a pair of handles **230**, a handle lock **202**, and mounting arms **210** for securely coupling to mounting tangs **19**, **29** of implant **100**, for example. Inserter **200** may further include a tightening knob **211** that is connected to a drive shaft **220** having a drive end **221**. Drive end **221** may have a size and shape generally corresponding to a size and shape of the various drive features of locking screw **50**, for example the internal threaded surface **52**, first drive feature **53a**, and/or second drive feature **53b**. In the example embodiment shown in FIG. **24**, drive end **221** includes an end portion having an outside threaded surface with a corresponding size and shape to the internal threaded surface **52** of locking screw **50**. In various embodiments, tightening knob **211** may rotate drive shaft **220** and drive end **221** to engage drive end **221** with locking screw **50** and pull implant **100** towards inserter tool **200** such that mounting tangs **19**, **29** are securely nested within corresponding channels of mounting arms **210**.

Additionally, an end user may rotate inserter **200** thereby translating a rotational force through drive end **221** to locking screw **50**, e.g., via first drive feature **53a**, and/or second drive feature **53b**.

(82) As seen best in FIGS. 23A and 23B, inserter **200** may include a pair of handles **230**, a stationary arm **252**, a primary pivoting arm **250**, and a secondary pivoting arm **251**. For example, to expand implant **100** an end user may toggle handle lock **202** to an unlocked position and push down on thumb indentation **231** of the handle **230** that is connected to the primary pivoting arm **250** and secondary pivoting arm **251**. In doing so, primary pivoting arm **250** may pivot with respect to medial pivot point **240** and secondary pivoting arm **251** may pivot with respect to distal pivot point **245**, for example. The path of travel of secondary pivoting arm **251** may lift up on the corresponding mounting tang **19** or **29** and cause the superior endplate **10** and inferior endplate **20** to separate from one another at the proximal end, for example. In doing so, an end user can lordose implant **100** to a desired angle. For example, as seen best in FIG. 24, secondary pivoting arm **251** has lifted up on mounting tang **19**, which is nested in a corresponding channel of mounting arm **210**.

(83) Once implant **100** is lordosed to a desired configuration, an end user may rotate drive shaft **220** and drive end **221** to tighten locking screw **50** as explained previously. After locking screw **50** is relatively tight, the end user may continue to apply a rotational force to locking screw **50** until a proximal portion comprising the cylindrical part having an internal threaded surface **52**, and the first drive feature **53a** breaks off at breakoff location **55**. For example, once locking screw **50** is tightened to a designed torque, the locking screw **50** may shear off as explained previously. At least one advantage of utilizing the locking screw **50**, is that it may prevent over tightening which can cause deformation to implant **100**. As shown in FIG. 25, implant **100** has been expanded to a desired position and/or lordotic angle. Additionally, locking screw **50** has locked the relative position of the superior endplate **10** with respect to the inferior endplate **20** and the proximal portion of locking screw **50** has been broken off as explained above.

(84) FIGS. 26 and 27 are various views of a surgical instrument **300** for use with disclosed expandable implants **100**. In some embodiments, surgical instrument **300** may be referred to as a breakoff instrument and may be used to breakoff the mounting tangs **19**, **29** of implant **100**. In the example embodiment, surgical instrument **300** comprises a first instrument **310** and a second instrument **320**. First instrument **310** may extend in a longitudinal direction from handle **312** towards gripping end **311**. Similarly, second instrument **320** may extend in a longitudinal direction from handle **322** to gripping end **321**. Gripping ends **311**, **321** may comprise a channel having a size and shape generally corresponding to mounting tangs **19**, **29**. For example, as seen best in FIG. 27, the mounting tangs **19**, **29** may be insert inside of the corresponding channels of gripping ends **311**, **321**. After the mounting tangs **19**, **29** are nested within gripping ends **311**, **321** an end user may push laterally outward and/or inward against handles **312**, **322** to breakoff the corresponding mounting tang **19**, **29**. For example, as shown in FIG. 28 expandable implant **100** is in an expanded and lordosed configuration and the proximal portion of locking screw **50** and tangs **19**, **29** have been broken off.

(85) FIG. 29 is a reference drawing showing the human spine of which various disclosed implant embodiments may be installed in. FIG. 30 is a reference drawing showing various planes and reference directions of which the various disclosed implant embodiments may move in or act in with reference to a patient **1**.

(86) Referring generally to FIGS. 31-37 a second implant **400** embodiment is disclosed. Implant **400** may include the same, similar, and/or substantially the same components and functionality as explained above with respect to implant **100**. Accordingly, duplicative description will be omitted. It shall be understood that various components and functionality of implant **100** are readily combinable with implant **400** and vice versa unless the context clearly indicates otherwise.

(87) FIG. 31 is a perspective view of a second implant **400** embodiment. In this embodiment, implant **400** extends in a proximal-to-distal direction between a proximal end **400P** and a distal end **400D** and extends in a width-wise direction between a first lateral end **400L** and a second lateral end **400L**. Additionally, implant **400** includes a superior endplate **410** and an inferior endplate **420**

having substantially similar features and functionality as explained above with respect to superior endplate **10** and inferior endplate **20** of implant **100**, for example. However, in this embodiment, superior endplate **410** may include a first gripping protrusion **419** extending in a proximal direction from the proximal end **400P** of the superior endplate **410**. Similarly, inferior endplate **420** may include a second gripping protrusion **429** extending in a proximal direction from the proximal end **400P** of the inferior endplate **420**. In this embodiment, a size and shape of first gripping protrusion **419** is substantially the same as a size and shape of the second gripping protrusion **429**. However, in other embodiments the first and second gripping protrusions **419**, **429** may be differently sized and shaped, e.g., to bias the implant towards a surgical instrument and/or orientation for insertion. In this embodiment, and in the closed position, each gripping protrusion **419**, **429** is disposed at approximately the same distance from an axis of rotation of breakoff set screw **450**. In this embodiment, gripping protrusions **419**, **429** may replace the need for the tangs **19**, **29** of implant **100**, for example. However, the concepts of utilizing breakoff tangs **19**, **29** and gripping protrusions **419**, **429** are not necessarily mutually exclusive and attributes of one may be combined and/or modified in view of the other.

(88) In various embodiments, gripping protrusions **419**, **429** may include various types of contouring to facilitate grasping of gripping protrusions **419**, **429** with a corresponding inserter, for example surface indentations, surface outdents, channeling, etc. In the example embodiment, gripping protrusion **419** comprises a superior gripping surface **419A** including an indented portion and an outdented chamfered portion at a proximal most end thereof, for example. Additionally, gripping protrusion **419** comprises an inferior gripping surface **419B** including an indented portion and an outdented chamfered portion at a proximal most end thereof, for example. Likewise, gripping protrusion **429** comprises a superior gripping surface **429A** including an indented portion and an outdented chamfered portion at a proximal most end thereof, for example. Additionally, gripping protrusion **429** comprises an inferior gripping surface **429B** including an indented portion and an outdented chamfered portion at a proximal most end thereof, for example. In this way, gripping protrusions **419** and **429** are shaped like dovetails and a corresponding inserter tool may comprise a corresponding shaped dovetail groove which may grasp onto and/or slip over gripping protrusions **419** and **429** (not illustrated).

(89) Referring generally to FIGS. **32**, **33**, and **34** various exploded parts views of implant **400** are illustrated. FIG. **32** is a first perspective exploded parts view of implant **400**, FIG. **33** is a second perspective exploded parts view of implant **400**, and FIG. **34** is a side exploded parts view of implant **400**. In the example embodiment, the superior and inferior endplates **410**, **420** of implant **400** may be hingedly coupled together by hinge member **440** and arcuate channel **412** having similar attributes as explained above with respect to hinge member **40** and channel **12** of implant **100**, for example. Additionally, superior endplate **410** may also include a core **415** having an aperture **415A** and inferior endplate **420** may include a core **425** having a threaded aperture **425A** having similar attributes to core **15** and core **25** as explained above with respect to implant **100**, for example. In the example embodiment, implant **400** utilizes a breakoff screw **450** for locking of a position of the superior and inferior endplate **410**, **420**. Breakoff screw **450** may include an external thread pattern **451** on an outside circumferential surface thereof, for example. The external thread pattern **451** of breakoff screw **450** may have a size and shape generally corresponding to the threaded aperture **425a** of core **425** of the inferior endplate **420**, for example. In various embodiments, an engagement surface **454** may be disposed adjacent and proximal of external thread pattern **451**. In the example embodiment, engagement surface **454** is shaped like a washer and is directly connected to breakoff screw **450**. However, in other embodiments, engagement surface **454** may be a washer or separated element, for example. In some embodiments, engagement surface **454** may be conically shaped. In the example embodiment, engagement surface **454** may include a relatively planar and/or flat distal surface and/or proximal surface.

(90) In various embodiments, a proximal end of breakoff screw **450** may include a first flexible

tang **452A** and a second flexible tang **452B** defining a discontinuous cylindrical shaped aperture **452** therebetween. Additionally, the first flexible tang **452A** and second flexible tang **452B** may each include an outdent at a proximal end thereof that is shaped like a segment of an annular ring. In the example embodiment, the first flexible tang **452A** may flex inward towards the second flexible tang **452B** under loading and vice versa due to the gap between them. At least one advantage of this configuration is that it may facilitate securing breakoff screw **450** to a corresponding drive tool (not illustrated) and the retention of the broken off part. For example, a drive tool may comprise a drive end having a female cavity with a corresponding size and shape to the drive features **453A**, **453B**. In various embodiments, the cavity may include a pair of indentations corresponding in size and shape to the outdents of flexible tangs **452A** and **452B**, for example. In use, an end user may align the flexible tangs **452A**, **452B** with the cavity, push down against the flexible tangs **452A**, **452B** which may cause them to flex inward towards each other such that they may slide within the cavity until the outdents of flexible tangs **45A** and **452B** are seated within the corresponding indents of the drive tool. Thereafter, an end user may continue to rotate and or tighten breakoff screw **450**. In this way, after a proximal portion of breakoff screw **450** is broken off it may remain retained by the inserter due to the flexible tangs **452A** and **452B** being seated within the corresponding indents.

(91) In some embodiments, aperture **452** may be understood as a cylindrical protrusion extending in a proximal direction and having a first slit and a second slit extending along the length thereof such that the cylindrical protrusion is compressible. In the example embodiment, a first drive feature **453A** and a second drive feature **453B** may be disposed adjacent to and distally with respect to a proximal most end of breakoff screw **450**. Additionally, first and second drive features **453A**, **453B** may be disposed proximally with respect to engagement surface **454**. In various embodiments, a breakoff location may be positioned between and/or adjacent to drive features **453A**, **453B**, which will be explained in further detail below. In the example embodiment, drive features **453A**, **453B** take a hexalobular shape, although various other shapes such as hexagonal, polygonal, torx, etc. are also contemplated. In some embodiments, a surgical drive tool having a corresponding socket may be coupled to drive features **453A** and/or **453B** to cause rotation of breakoff screw **450**. Similarly as explained above with respect to locking screw **50**, once breakoff screw **450** has been sufficiently tightened a proximal portion may breakoff and/or shear off while the distal portion may remain coupled to implant **400** locking a relative orientation of superior endplate **410** and inferior endplate **420** in place.

(92) As seen best in FIGS. **35-37**, breakoff screw **450** may extend in a longitudinal direction along a longitudinal axis L-A that is coaxially aligned with breakoff screw **450**. FIG. **35** is a first side view of breakoff screw **450** in which the superior surface **452A** and inferior surface **452B** of discontinuous aperture **452** are visible. FIG. **36** is a second side view of a breakoff screw **450** that is rotated about 90 degrees with respect to FIG. **35** in which only the superior surface **452A** is visible. With reference to FIG. **37**, in various embodiments, breakoff location **455** may comprise a recessed fracture surface F-S that is inset with respect to a leading edge (proximal most edge) of drive feature **453A**. At least one advantage of the recessed fracture surface may be that delicate tissue is prevented and/or suppressed from coming into contact with relatively sharp ends of the fracture surface. In the example illustration, a relative location of the recessed fracture surface is represented by dashed lines F-S. In the example embodiment, breakoff location **455** may be considered as the boundary between a proximal portion **450A** and a distal portion **450B** of breakoff screw **450**, for example. In this embodiment, the boundary between drive features **453A** and **453B** comprises a necked down portion **458** extending from a distal end of drive feature **453A** to an inset portion of drive feature **453B** that is inset with respect to an outermost and/or proximal most surface of drive feature **453B** thereby defining a portion of breakoff set screw **450** having a minimum cross sectional diameter. Accordingly, when breakoff screw **450** is sufficiently tightened within threaded aperture **425A** of core **425** such that the breakoff location **455** experiences a

sufficient torque the proximal portion **450A** may breakoff from the distal portion **450B**. For example, when a sufficient rotational force is applied to the proximal end of breakoff screw **450** while a distal end of breakoff screw **450** is stationary, i.e., when breakoff screw **450** is secured in a locked position and a continued rotational force (torque) is applied to the proximal end of breakoff screw **450** the drive feature **453A** and cylindrical end having the discontinuous aperture **452** may breakoff. For further explanation in the similar context of implant **100**, see FIGS. **10** and **11** and the corresponding discussion thereof.

(93) Referring to FIGS. **38-42** an example swaging process is performed to a distal most end of breakoff screw **450**. FIG. **38** is a perspective view of a swaging fixture **500**, and FIG. **39** is a cross section view of a swage mandrel and a distal end **450D** of a breakoff screw **450** before the commencement of a swage process. FIG. **40** is a cross section view showing a result of a swage process, and FIG. **41** is an enlarged view of region S-W of FIG. **40**. In the example embodiment, swaging fixture **500** comprises a swaging ram **501** and a swaging mandrel **503** that are supported by the base of the apparatus. The swaging mandrel **503** may include an outdent that corresponds to and is slightly larger than the distal most indent **498** (swage bore) of breakoff screw **450**, for example. As seen in region S-W of FIG. **40**, when swaging mandrel **503** is advanced into the distal most indent **498** (swage bore) a flared out portion **499** (swaged portion) is formed at a distal most end of breakoff screw **450**. An example advantage of a swaged end may be that it facilitates retention of the breakoff screw **450** such that it serves as a stopping structure preventing breakoff screw **450** from backing out of implant **100**.

(94) It shall be understood that although breakoff screw **450** is explained concurrently with implant **400** and in the context of an intervertebral implant, the concepts of breakoff screw **450** may be applied to other embodiments used for alternate purposes, for example for use in a pedicle screw to insert in a tulip to tighten down a rod. It should be understood that various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and accompanying drawings. For example, features, functionality, and components from one embodiment may be combined with another embodiment and vice versa unless the context clearly indicates otherwise. Similarly, features, functionality, and components may be omitted unless the context clearly indicates otherwise. It should also be understood that, depending on the example, certain acts or events of any of the processes or methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques).

(95) Referring generally to FIGS. **42-51**, a third implant **400Z** and a surgical tool **600** for use with various implants is disclosed. FIGS. **42A-42B** illustrate a third embodiment of an implant **400Z**. Implant **400Z** may have the same, similar, and/or substantially the same components and functionality as explained above with respect to implant **400**, for example. Accordingly, duplicative description will be omitted. A difference may be that breakoff screw **450** does not include the discontinuous cylindrical shaped aperture **452** (see FIG. **34**) having tangs **452A**, **452B**, for example. Rather, breakoff screw **450**, to the extent included, may include a continuous cylindrical shaped aperture **452Z** without the discontinuity. In some embodiments, a similar screw may be used that is not a breakoff screw but rather may be referred to as a locking screw. Additionally, in this embodiment, implant **400Z** may include a first gripping indentation **419Z** and a second gripping indentation **429Z** (see FIG. **42A**) in lieu of the gripping protrusions **419**, **429** (as shown in FIG. **33**), for example. In various embodiments, first gripping indentation **419Z** may be formed as a part of the superior endplate **410** and second gripping indentation **429Z** may be formed as a part of the inferior endplate **420**, for example.

(96) In various embodiments, gripping indentations **419Z**, **429Z** may include various types of contouring to facilitate coupling with a corresponding surgical tool **600**. In this embodiment, gripping indentations **419Z**, **429Z** each comprise a slotted indentation extending in a proximal to distal direction with a curved superior surface **419S**, **429S** and a curved inferior surface **419I**,

4291. Additionally, gripping indentations **419Z**, **429Z** each have an open channel portion **419X**, **429X** adjacent the breakoff screw **450**, to the extent a breakoff screw is included. As will be explained in further detail below, corresponding gripping protrusions of a surgical tool **600** may have a substantially similar size and shape to the gripping indentations **419Z**, **429Z**, for example. (97) FIGS. **43A-43B** illustrate a similar embodiment as FIGS. **42A-42B**. As shown, the implant **400Z** includes angled engagement features **14**. In the example embodiment, engagement features **14** extend diagonally across the exposed uppermost surface of superior endplate **410** and across the exposed lowermost surface of inferior endplate **420**. The engagement features **14** comprise flattened top rails that are sequentially spaced apart with rounded bottom valleys **14V** therebetween. As seen best in the top down view of FIG. **43B**, the engagement features **14** of the superior endplate **410** are oriented at an angle β with respect a proximal face **410F** of the superior endplate **410**. Similarly, the engagement features **14** of the inferior endplate **420** are oriented at an angle β with respect a proximal face **420F** of the inferior endplate **420** (see FIG. **43A**). In various embodiments, the angle β may be about 20-40 degrees and in at least one embodiment the angle β may be angled at about 30 degrees. At least one advantage of this orientation may be a relatively greater resistance and/or suppression of expulsion of the implant **400Z** relative to embodiments in which engagement features solely extend horizontally across the implant **400Z**. In the example embodiment of FIGS. **43A-43B**, by orienting the engagement features **14** diagonally, the implant **400Z** may resist expulsion in multiple directions, e.g., forward flexion/extension and lateral bending.

(98) Referring generally to FIGS. **44-48**, various views of a surgical tool **600** are disclosed. FIG. **44** illustrates the surgical tool **600** coupled to implant **400Z**. FIG. **44** is a perspective view of surgical tool **600**; FIG. **45** is a first side view of surgical tool **600**; and FIG. **46A** is a second side view of surgical tool **600**. FIG. **46B** is a third side view of surgical tool **600** in an operative position and FIG. **46C** is a perspective view of surgical tool **600** in the operative position of FIG. **46B**. FIG. **47A** is an exploded parts view of surgical tool **600** with select parts removed for ease of understanding and FIG. **47B** is an exploded parts view of surgical tool **600** showing additional detail and parts. FIG. **48** illustrates surgical tool **600** immediately prior to coupling with implant **400Z**. In the example embodiment, tool **600** may serve several purposes. For example, tool **600** may be used to insert implant **400Z**, expand implant **400Z**, and in some embodiments may also rotate breakoff screw **450** to cause it to separate into two portions along breakoff location **455** (see FIG. **34**).

(99) With reference to the perspective view of FIG. **44** and the exploded parts views of FIGS. **47A** and **47B**, tool **600** may include a gripping handle **601** that is securely coupled to first body portion **607**. First body portion **607** may rotatably support an inner shaft **606** and an outer shaft **602** therein. Shafts **606**, **602** may extend through aperture **607A** of first body portion **607** in a longitudinal direction from a proximal end to a distal end. The inner shaft **606** may be securely connected to turn knob **603** at a proximal end **600P** and the outer shaft **602** may be securely connected to turn handle **604** at the proximal end **600P**. The inner shaft **606** may extend through outer shaft **602** and include a thread pattern **606T** at a distal end thereof for securely connecting to corresponding threads **456** of breakoff screw **450** (see FIG. **42B** for example). The inner shaft **606** may be independently rotatable relative to outer shaft **602** by rotating turn knob **603**. Additionally, the outer shaft **602** may be independently rotatable relative to inner shaft **606** by rotating turn handle **604**, for example. Generally, both the inner shaft **606** and outer shaft **602** will interact with and/or couple to breakoff screw **450** as will be explained in further detail below.

(100) With reference to FIGS. **45** and **46A-46C**, tool **600** may include an actuator **605** in the form of a trigger that is connected to first body portion **607** at pin **605C**. Actuator **605** may be actionable to expand implant **400Z** by pivoting a linkage assembly **608** including first arm **610** and second arm **611** relative to a third fixed arm **620**, for example. Second arm **611** may be coupled to actuator **605** via pin **605B** such that pin **605B** may be linearly translatable forward and backward in a proximal-to-distal direction within slotted aperture **605A**. In turn, second arm **611** may be coupled

to first arm **610** via pin **610B** such that second arm **611** may pivot up and down. Additionally, first arm **610** may be coupled to third arm **620** and in various embodiments third arm **620** may be fixed relative to first body portion **607**. Additionally, third arm **620** may include a protrusion **620B** which may be slidably seated within a curved vertical slot, for example slot **610A**, and thereby guide motion of arm **610** as it pivots up and down. As seen best in FIGS. **46B** and **46C**, depressing actuator **605** may cause linkage assembly **608** to pivot by causing second arm **611** to move distally such that first arm **610** pivots relative to third arm **620** and implant **400Z** may be expanded.

(101) With reference back to FIG. **44**, third arm **620** may include an aperture extending therethrough in a proximal to distal direction for accommodating a linearly translatable shaft **621**. Shaft **621** may independently move forward and backward in a proximal to distal direction within the through aperture **624** of third fixed arm **620** (see FIGS. **47A** and **47B**), for example. Shaft **621** may include a charging handle at a proximal end thereof allowing an end user to manipulate shaft **621** in a forward and backward motion (proximal motion and distal motion in the proximal-to-distal direction). In some embodiments, an end user may translate charging handle **622** rotatably in a clockwise or counterclockwise motion, for example to lock shaft **621** in a relative position. Additionally, in some embodiments, when shaft **621** is not in a locked position, depressing actuator **605** will not expand implant **400Z** for reasons explained in more detail below.

(102) With reference to FIG. **48**, first arm **610** may include a gripping outdent **619** having a size and shape generally corresponding to a size and shape of gripping indentation **419Z**. Additionally, first arm **610** may include a counter torque surface **615** having a size and shape generally corresponding to a size and shape of the proximal surface of the superior endplate **410**, for example. Similarly, shaft **621**, extending through fixed arm **620**, may include a gripping outdent **629** having a size and shape generally corresponding to a size and shape of gripping indentation **429Z** (as shown in FIG. **42A**). Additionally, second arm **620** may include a counter torque surface **625** having a size and shape generally corresponding to a size and shape of the proximal surface of the inferior endplate **420**, for example. In this way, when shaft **621** is moved distally such that the outdent **629** is seated within indent **429Z** and outdent **619** is seated within indentation **419Z** (as shown in FIG. **42A**) an end user may activate actuator **605** to expand implant **400Z** while surfaces **615**, **625** resist a twisting motion of implant **100**. For example, as shown in FIGS. **46B** and **46C** an end user may depress actuator **605** by squeezing the trigger which in turn may cause the linkage assembly **608** to pivot the first arm **610** relative to fixed arm **620** such that tool **600** may urge the superior endplate **410** and inferior endplate **420** away from one another.

(103) With reference to the cross-section drawings of FIGS. **49**, **50**, and **51** additional features and functionality of tool **600** will be explained. In FIGS. **49** and **50**, it is shown that the threaded end of inner shaft **606** is threadably engaged with the threaded aperture **456** of breakoff screw **450**.

Additionally, a proximal drive feature **453A** is mated within a corresponding female drive feature **602A** (see FIG. **48**) at the distal most end of outer shaft **602**. As explained previously, inner shaft **606** and outer shaft **602** are independently operable. In use, an end user may initially position inner shaft **606** adjacent to/within the front portion of threaded aperture **452** and rotate inner shaft **606** via turn knob **603**. As inner shaft **606** is rotated, implant **400Z** is pulled towards tool **600**. FIGS. **49-51** show implant **400** after inner shaft **606** has been sufficiently rotated such that implant **400Z** abuts the counter torque surfaces **615**, **625** and the gripping protrusion **619** may be seated within the corresponding gripping indentation **419Z**. With reference to FIG. **51**, once implant **400Z** is pulled sufficiently towards tool **600**, an end user may actuate shaft **621** by pushing on charging handle **622** to overcome the biasing force of spring **623** such that gripping protrusion **629** may be seated within the corresponding gripping indentation **429Z**, for example. In some embodiments, an end user may rotate charging handle **622** such that biasing force of spring **623** is prevented from urging shaft **621** away from implant **400Z**. At this point, the implant **400Z** may be in an operatively engaged position with tool **600**.

(104) Once implant **400Z** is in the operatively engaged position, an end user may insert implant

400Z into a disc space between a superior vertebra and inferior vertebra (cephalad vertebra and caudal vertebra), for example. After the implant **400Z** is positioned between the superior and inferior vertebrae, an end user may depress actuator **605** thereby causing the linkage assembly **608** to pivot and spread the superior endplate **410** apart from the inferior endplate **420**. Once the end user has expanded implant **400Z** to an appropriate position, the end user may rotate turn handle **604** which will rotate breakoff screw **450** via first drive feature **453A**. The end user may continue to rotate breakoff screw **450** such that it advances into implant **400Z** in a proximal to distal direction and locks implant **400Z** in the expanded configuration similarly as explained above. The end user may continue to rotate turn handle **604** until the torque applied to breakoff screw **450** is great enough that the breakoff screw **450** will shear into two pieces similarly as explained above. Notably, the broken off or sheared off portion of breakoff screw **450** may be retained by tool **600** on account of inner shaft **606** being threadably coupled to threaded aperture **452**.

(105) Unless otherwise specifically defined herein, all terms are to be given their broadest possible interpretation including meanings implied from the specification as well as meanings understood by those skilled in the art and/or as defined in dictionaries, treatises, etc. It must also be noted that, as used in the specification and the appended claims, the singular forms “a,” “an” and “the” include plural referents unless otherwise specified, and that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof

Claims

1. An expandable implant movable between a contracted position and an expanded position, comprising: an expandable body extending from a proximal end to a distal end in a proximal-to-distal direction and extending from a first lateral side to a second lateral side in a widthwise direction, the expandable body being defined by a superior endplate and an inferior endplate that are hingedly connected; the superior endplate comprising a first core having a screw slot and a distal engagement surface; the inferior endplate comprising a second core having a proximal engagement surface; and a locking screw extending in a longitudinal direction from a proximal end to a distal end, being extendable through the first core and the second core, and being movable between a locked position and an unlocked position, wherein, in the locked position, the locking screw urges the distal engagement surface of the first core against the proximal engagement surface of the second core, wherein the proximal end of the locking screw is defined by an exposed aperture having an internal threaded surface, and wherein the locking screw comprises a proximal portion and a distal portion that are configured to separate along a recessed fracture surface, and, when separated, the recessed fracture surface is recessed relative to a sidewall of the distal portion and the fracture surface is recessed relative to a proximal most edge of the distal portion.
2. The expandable implant of claim 1, wherein: the superior endplate comprises a first gripping indentation located at a proximal end thereof; and the inferior endplate comprises a second gripping indentation located at a proximal end thereof.
3. The expandable implant of claim 2, wherein: the first gripping indentation comprises a slot having a superior curved surface and an inferior curved surface; and the second gripping indentation comprises a slot having a superior curved surface and an inferior curved surface.
4. The expandable implant of claim 1, wherein: the superior endplate comprises a first plurality of engagement features that are angled 20 degrees to 40 degrees with respect to a proximal surface of the superior endplate; and the inferior endplate comprises a second plurality of engagement features that are angled 20 degrees to 40 degrees with respect to a proximal surface of the inferior endplate.
5. The expandable implant of claim 1, wherein: the superior endplate further comprises a channel

located adjacent the distal end and extending in the widthwise direction; and the inferior endplate further comprises a rail located adjacent the distal end and extending in the widthwise direction, the rail having a size and shape generally corresponding to a size and shape of the channel and being located within the channel.

6. The expandable implant of claim 5, wherein: the distal engagement surface of the first core comprises a first curved surface; the proximal engagement surface of the second core comprises a second curved surface; and the first curved surface is defined by a radius of a circle having a center point that is offset from an axis of rotation of the rail.

7. The expandable implant of claim 5, wherein: the superior endplate comprises at least one hook portion adjacent the channel; the inferior endplate comprises at least one relief portion adjacent the rail; the at least one hook portion has a size and shape generally corresponding to a size and shape of the at least one relief portion; and the at least one hook portion is disposed within the at least one relief portion.

8. The expandable implant of claim 7, wherein, in a cross-section view, the channel comprises an arcuate shape and the rail comprises an arcuate shape.

9. The expandable implant of claim 1, wherein: the locking screw comprises a threaded distal end and a drive feature; the first core is disposed proximally with respect to the second core; and the locking screw is threadably engaged with at least one of the first core and second core.

10. A system, comprising: an expandable implant movable between a contracted position and an expanded position, the implant comprising: an expandable body extending from a proximal end to a distal end in a proximal-to-distal direction and extending from a first lateral side to a second lateral side in a widthwise direction, the expandable body being defined by a superior endplate and an inferior endplate that are hingedly connected; the superior endplate comprising a first core having screw slot and a distal engagement surface; the inferior endplate comprising a second core having a proximal engagement surface; and a locking screw movable between a locked position and an unlocked position; wherein, in the locked position, the locking screw urges the distal engagement surface of the first core against the proximal engagement surface of the second core; and a surgical tool configured to move the implant from the contracted position to the expanded position and to move the locking screw between the locked position and the unlocked position, the surgical tool being capable of moving the locking screw into the locked position while supporting the implant in the expanded position, wherein the surgical tool further comprises: a body portion having an aperture extending therethrough; a pivotal linkage assembly including a first arm and a second arm; a third arm that is fixed relative to the body portion; an outer shaft extending through the body portion and having a drive feature at a distal end thereof for rotating the locking screw from the unlocked position to the locked position; and an inner shaft extending through the outer shaft and having a first thread pattern at an end thereof for drawing the implant towards the surgical tool.

11. The system of claim 10, wherein: the superior endplate comprises a first gripping indentation located at a proximal end thereof; the inferior endplate comprises a second gripping indentation located at a proximal end thereof; the surgical tool comprises a first gripping protrusion having a size and shape generally corresponding to a size and shape of the first gripping indentation; and the surgical tool comprises a second gripping protrusion having a size and shape generally corresponding to a size and shape of the second gripping indentation.

12. The system of claim 11, wherein: the first gripping indentation comprises a slot having a superior curved surface and an inferior curved surface; and the second gripping indentation comprises a slot having a superior curved surface and an inferior curved surface.

13. The system of claim 10, wherein the surgical tool comprises a pivotal linkage assembly including a first arm and a second arm.

14. The system of claim 10, wherein: the surgical tool is configured to separate the locking screw into the proximal portion and distal portion while the implant is in the expanded position and retain

the proximal portion of the locking screw via the inner shaft.

15. The system of claim 10, wherein the surgical tool further comprises an actuator for causing the pivotal linkage assembly to pivot relative to the third arm and thereby expand the implant.

16. The system of claim 10, wherein: the superior endplate further comprises a channel located adjacent the distal end and extending in the widthwise direction; and the inferior endplate further comprises a rail located adjacent the distal end and extending in the widthwise direction, the rail having a size and shape generally corresponding to a size and shape of the channel and being located within the channel.

17. The system of claim 16, wherein: the distal engagement surface of the first core comprises a first curved surface; the proximal engagement surface of the second core comprises a second curved surface; and the first curved surface is defined by a radius of a circle having a center point that is offset from an axis of rotation of the rail.
